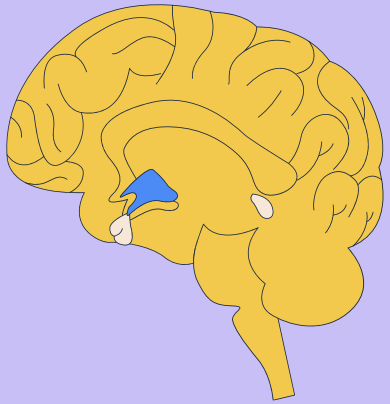




Principles of Hormone Testing

&

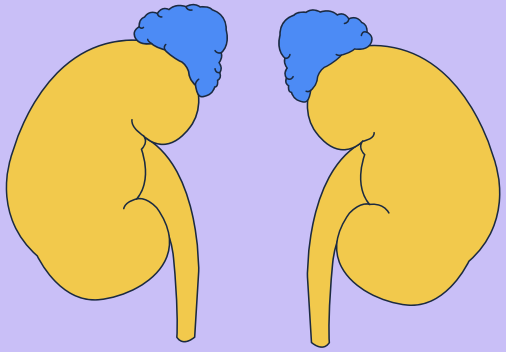
Laboratory Tests for Assessment of Adrenal Disorders

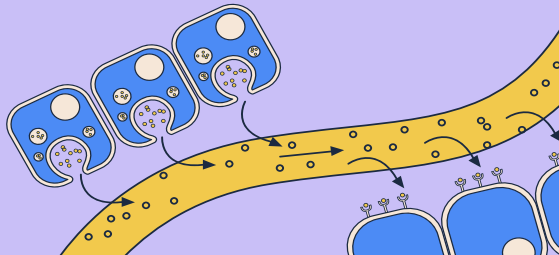
Assoc. Prof. Dr. Wilaiwan Sriwimol

Department of Pathology, Faculty of Medicine

Prince of Songkla University

(sriwimol.wi@gmail.com)





Learning Objectives

After today's lecture, you should be able to:



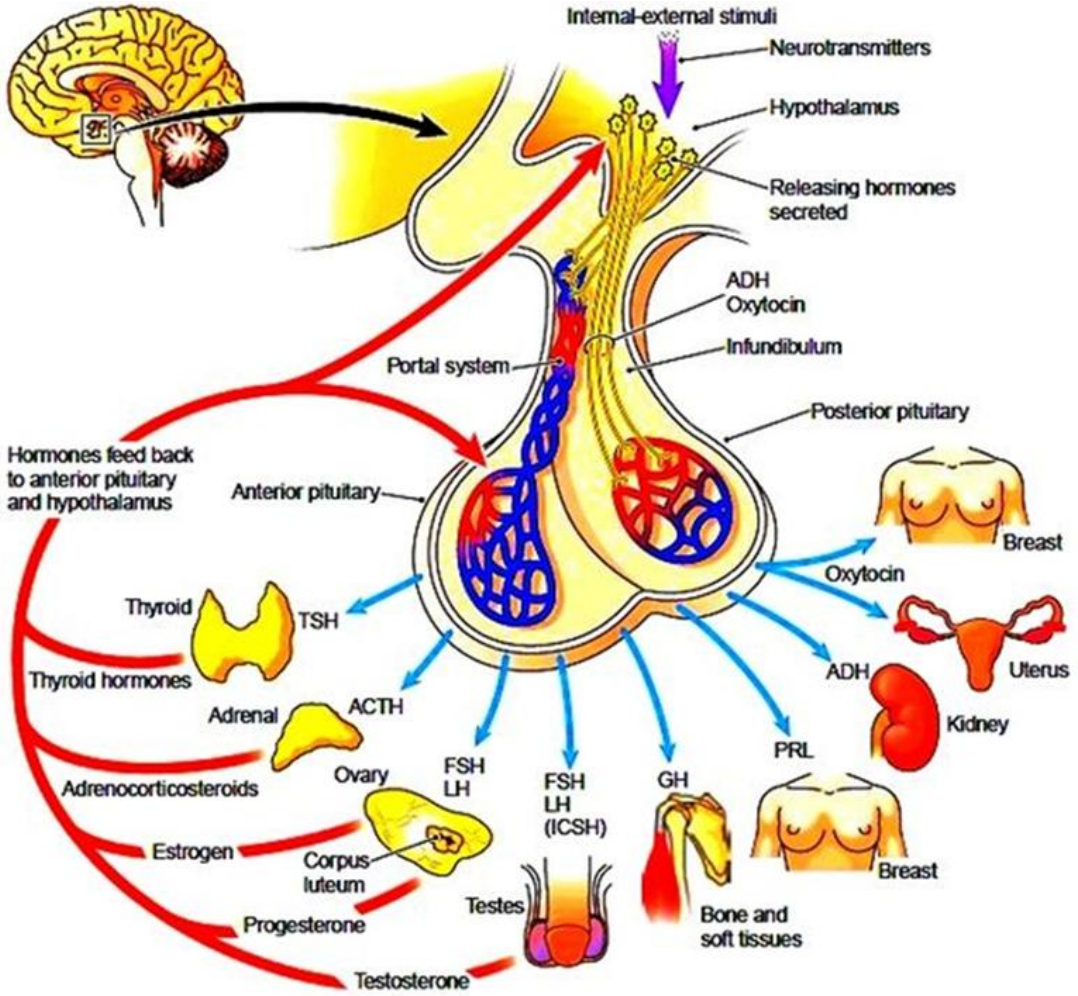
- ✓ **Explain** the principles of laboratory tests for hormone measurements
- ✓ **Select** the laboratory tests and **Interpret** the results for assessments of adrenal disorders

Outlines

Principles of hormone (endocrine) testing

Adrenal cortex: Laboratory investigations & Interpretations

Adrenal medulla: Laboratory investigations & Interpretations



Part I

Principles of Hormone (Endocrine) Testing

Endocrine Abnormalities

Hormone Excess (Hyperfunction)

- Hyperpituitarism
- Cushing's syndrome
- Conn's syndrome
- Pheochromocytoma



Hormone Deficiency (Hypofunction)

- Hypopituitarism
- Adrenal insufficiency (adrenocortical insufficiency)



Suspected abnormal/disorder

Select "laboratory tests" for hormone measurement

Indications/objectives
Precautions
Procedure
Interpretation



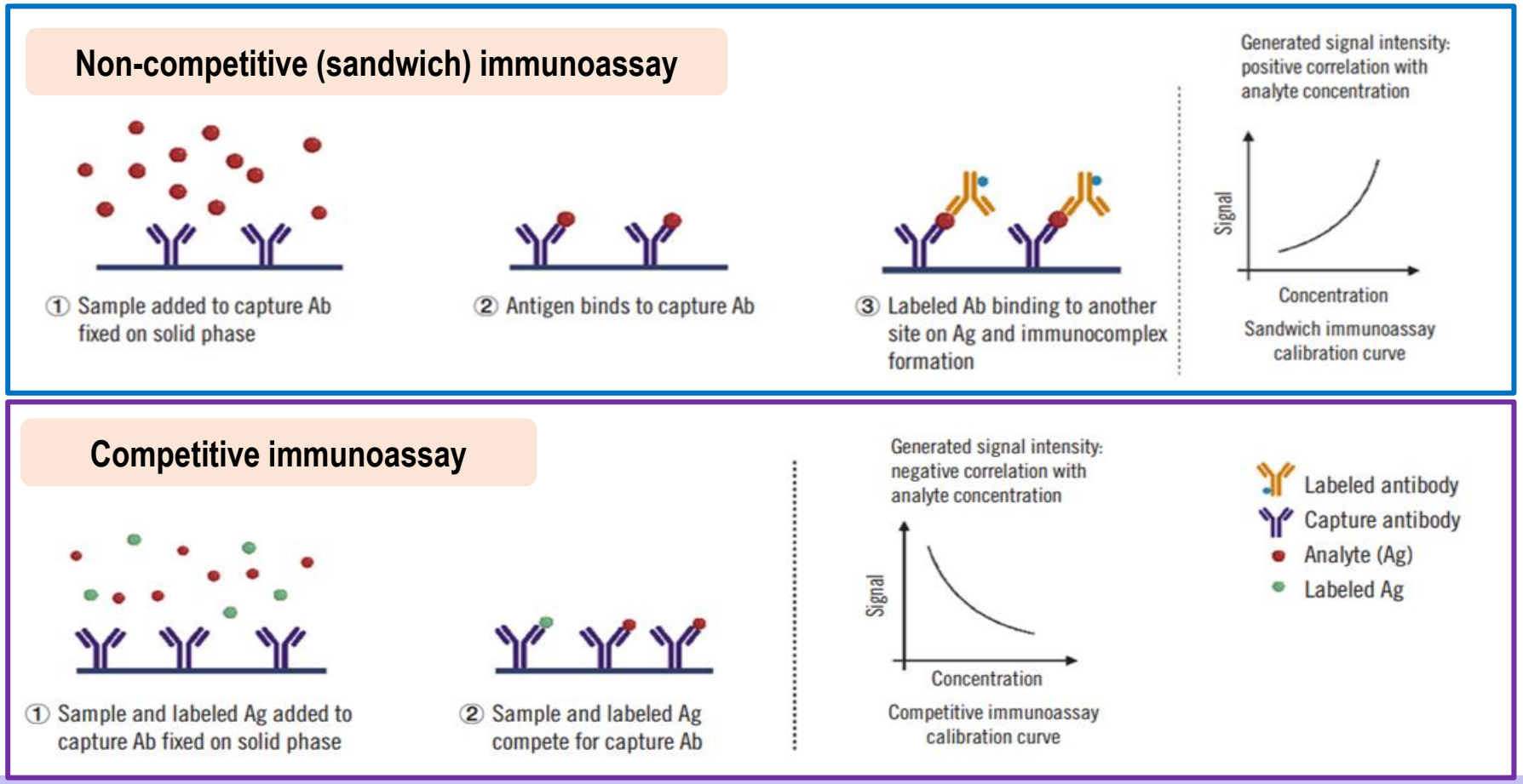






Measurements of Hormones and Related Analytes

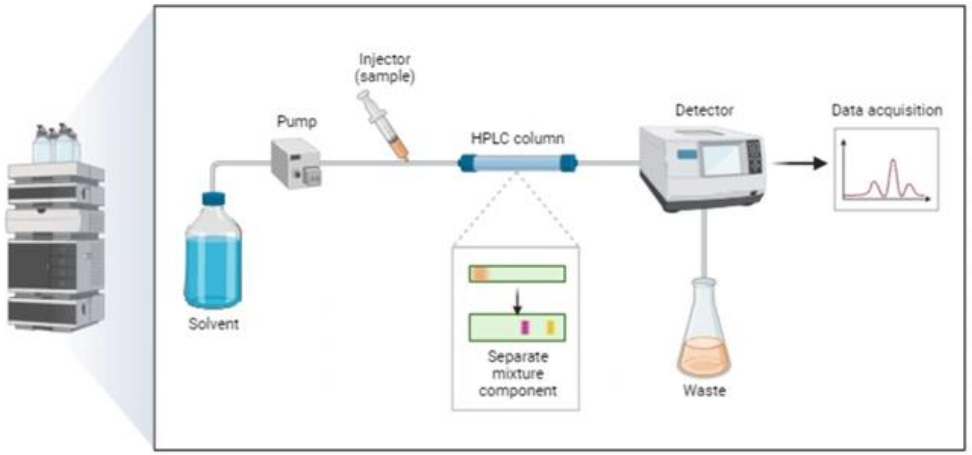
- Analytical methods → Immunoassay, HPLC, Liquid chromatography-Tandem mass spectrometry (LC-MS/MS)
- Most commonly used → Immunoassay → Competitive and non-competitive immunoassays

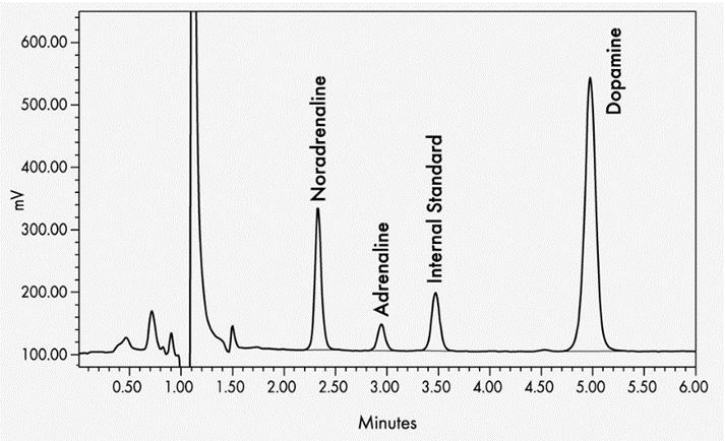


Ghazal K, et al. Ann Lab Med 2022;42:3-23.

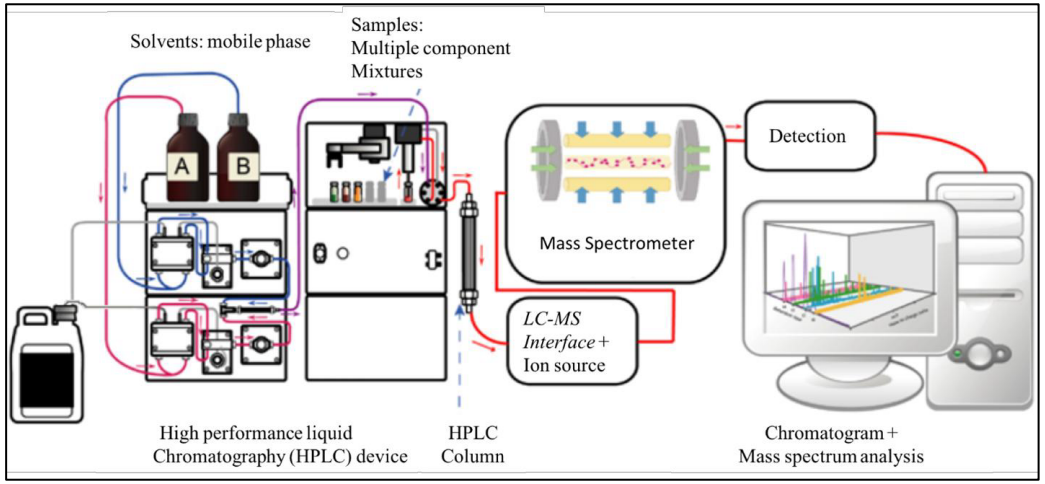
Measurements of Hormones and Related Analytes

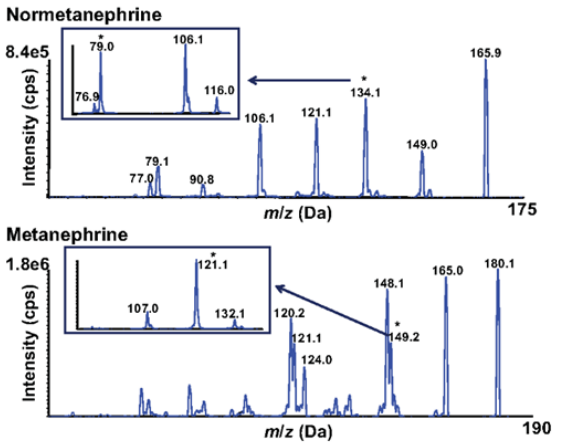
High-Performance Liquid chromatography (HPLC)

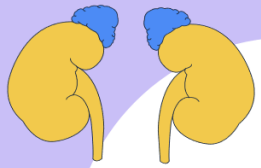




Tandem Mass Spectrophotometry *e.g.* LC-MS/MS







Consideration for Hormone Measurement & Interpretation

Pre-analytical conditions

- Stress, acute illness
- Medication
- Specimen collection
  - Circadian rhythm (time)
  - Body position/posture
  - Types of specimen
  - Types of anti-coagulants
  - Temperature for transportation
  - Patient preparation, *etc*.

Analytical pitfalls/limitations

- Assay interferences
  - Heterophile Ab
  - Human anti-animal Ab
  - Rheumatoid factors
  - Anti-analyte Ab
  - Drug interference
  - Biotin, Unknown interference
- Cross-reaction
- Macro-form of hormone
- Hook effect

Post-analytical phase

- Reference range
- Clinical decision limit/cut-off
- Biological factors
  - age and sex
  - BMI
  - Metabolic or concomitant diseases, *etc*.

More reading

- Ghazal K, *et al*. Hormone immunoassay interference: a 2021 update. Ann Lab Med 2022;42:3-23.
- Haddad RA, *et al*. Interpretation of common endocrine laboratory tests: technical pitfalls, their mechanisms and practical considerations. Clin Diabetes Endocrinol 2019;5:12.
- Schilbach K, *et al*. Pitfalls in the lab assessment of hypopituitarism. Rev Endocr Metab Dis 2024.

Hormone Measurement

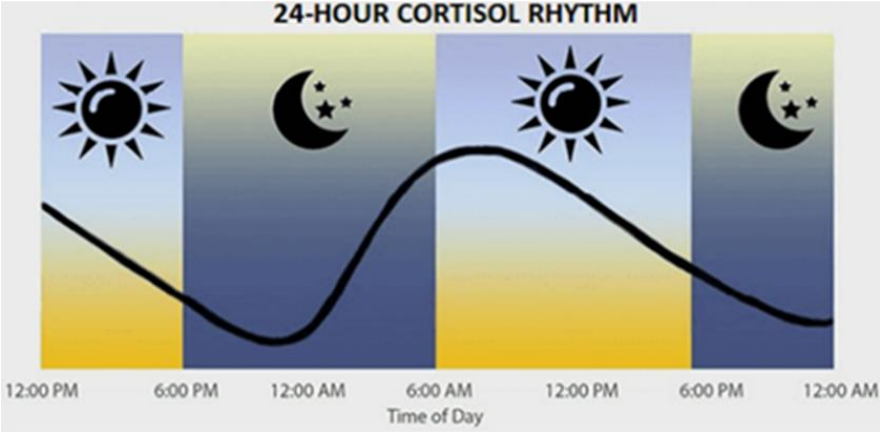
1. Direct measurement: Basal level

➢ *Plasma/serum hormone levels*

→ at 08:00 – 09:00 h for blood collection

→ at 23:00-24:00 h for blood collection

→ To check the diurnal variation

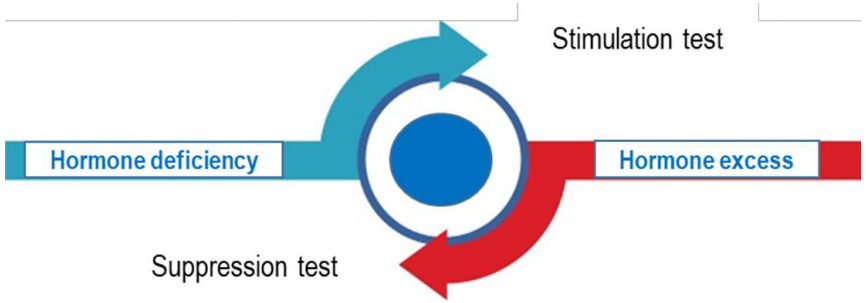


Test Subjected to diurnal variation
Acid phosphatases*
ACTH
Cortisol (& other adrenal steroid)
Growth hormone
Insulin
Parathyroid hormone
Renin/aldosterone
TSH*
Testosterone*
*higher in afternoon and evening
All others higher in the morning

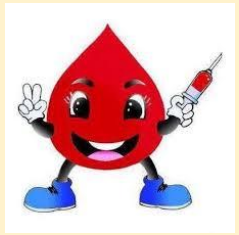
➢ *Hormone measurement in urine/saliva* → the free form of hormone, or the excretion of hormone/its metabolite

Hormone Measurement

2. Dynamic tests



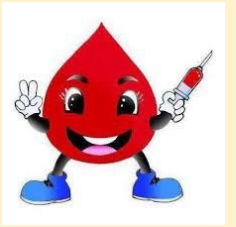
Procedure



Basal hormone measurement (before administration)

Administration of an "appropriate agent"

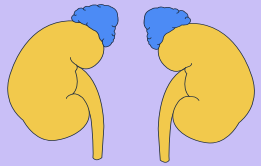
Synthetic form of hormone



Hormone measurement after administration at…….

Interpretation

- *Suppression tests:*
  Suppression → hormone concentration "After < Before"
- *Stimulation tests:*
  Response → hormone concentration "After > Before"



Part II

Disorders of Adrenal Cortex & Laboratory Investigations

- **Cortisol**: *Blood & Urine & Saliva*
- Plasma ACTH
- **Dynamic Tests**: Dexamethasone Suppression Test (DST)

  ACTH Stimulation Test

  CRH Stimulation Test

- Plasma Aldosterone, Plasma renin
- Aldosterone to renin ratio (ARR)

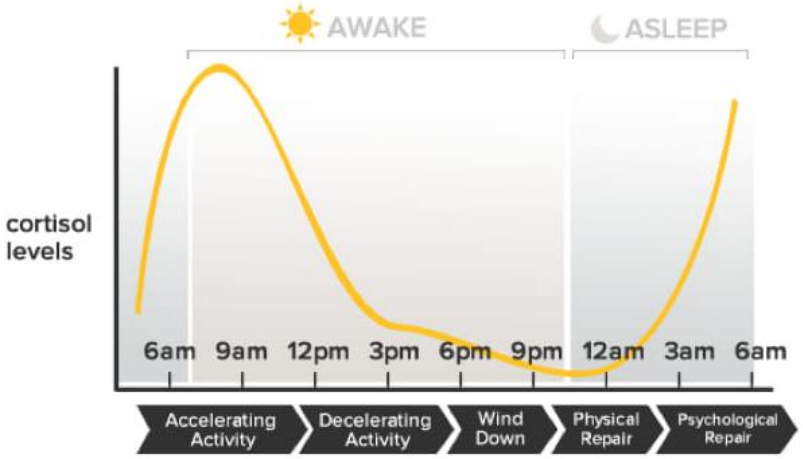
Adrenal hypofunction

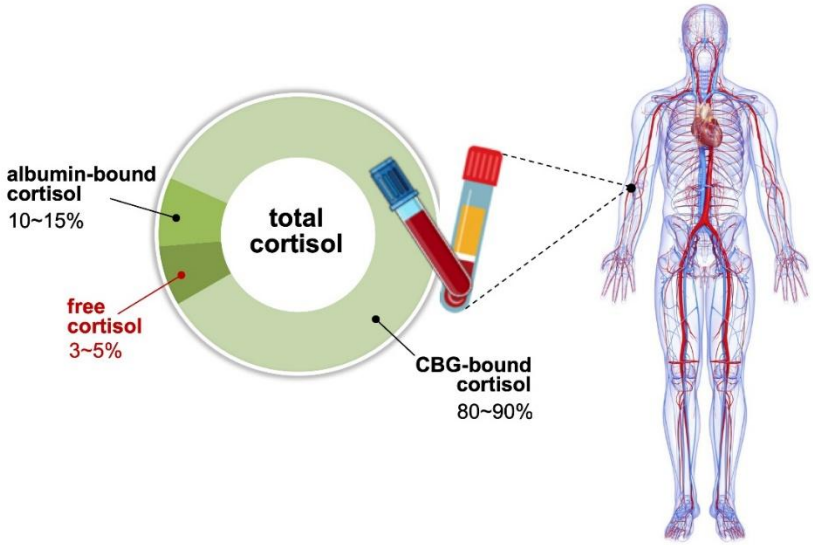
- Adrenal Insufficiency

Adrenal hyperfunction

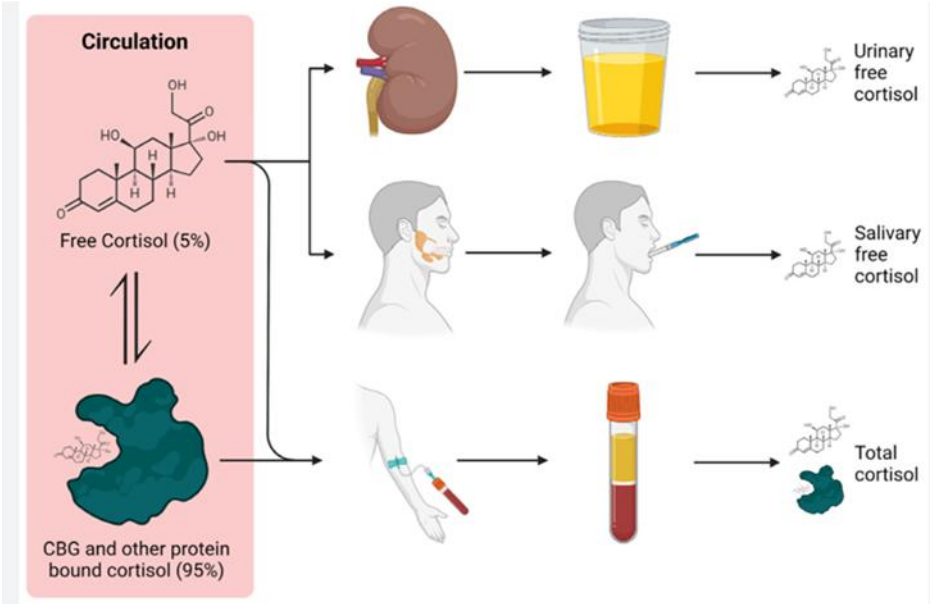
- Cushing's Syndrome
- Conn's Syndrome

Cortisol Measurement





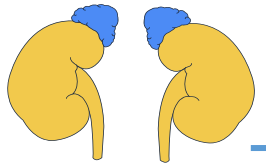
- *Blood* (Serum or heparinized plasma)
  - collected at 08:00 – 09:00 h
  - collected at 23:00-24:00 h (midnight)
    → to detect loss of diurnal variation
  - collected at specific times (dynamic tests)



- *Urine*
  - 24-h urine (UFC)

- *Saliva*
  - late-night/midnight saliva

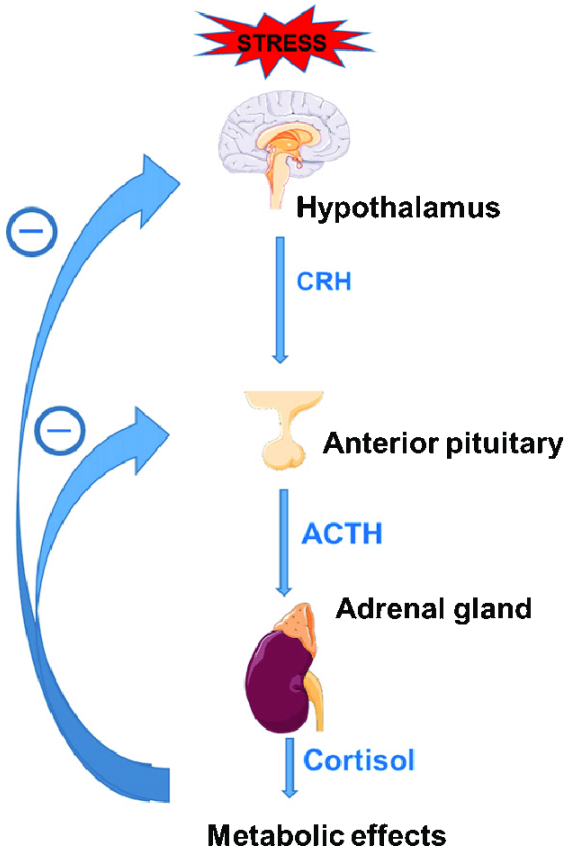




Cortisol Measurement

	Serum/Plasma	Urine	Saliva
Test	Serum/Plasma cortisol	24-h Urinary free cortisol	Late night or midnight saliva
Measurement	Total cortisol	Free cortisol	Free cortisol
Time to collect specimen	- Morning (08:00-09.00 A.M.) - Specific times	24-h urine collection	Late night or midnight
Advantages	Convenience specimen collection	■ No effects from circadian variation and the CBG conc. ■ Non-invasive specimen	■ Non-invasive specimen ■ No effects from the CBG conc.
Limitations	■ Circadian variation ■ Influenced by CBG concentration - Increased CBG conc. (e.g. estrogen, pregnancy, oral contraceptives, obesity, hyperglycemia, *etc.*) >> False positive of plasma/serum cortisol - Decreased CBG conc. (e.g. severe illness, hyperthyroidism, chronic liver disease, nephrotic syndrome, starvation, *etc.*) >> False negative of plasma/serum cortisol	■ False positive - e.g. stress, proteinuria, polyuria, *etc.* ■ False negative - e.g. incomplete 24-h urine collection, chronic kidney disease (eGFR <60 mL/min), urinary tract infection (UTI)	■ False positive - e.g. stress, sleep, deprivation, shift work, alcoholism, smoking, liquarice (ชะเอม), grapefruit juice, *etc.* ■ False negative - Sjogren's syndrome ■ Inappropriate for people with oral/dental diseases (e.g. ulcers)

Plasma ACTH



- ✓ To detect, diagnose and monitor conditions associated with excessive (hyper-) or deficient (hypo-) cortisol
- ✓ Differentiating between causes of Cushing's syndrome (ACTH-dependent *vs* ACTH-independent)

Increased Plasma ACTH

- Addison's disease
- Cushing's disease
- Ectopic ACTH-producing tumors
- Congenital adrenal hyperplasia (CAH)
- *Others*; stress, hypoglycemia, pregnancy, some medications, *etc.*

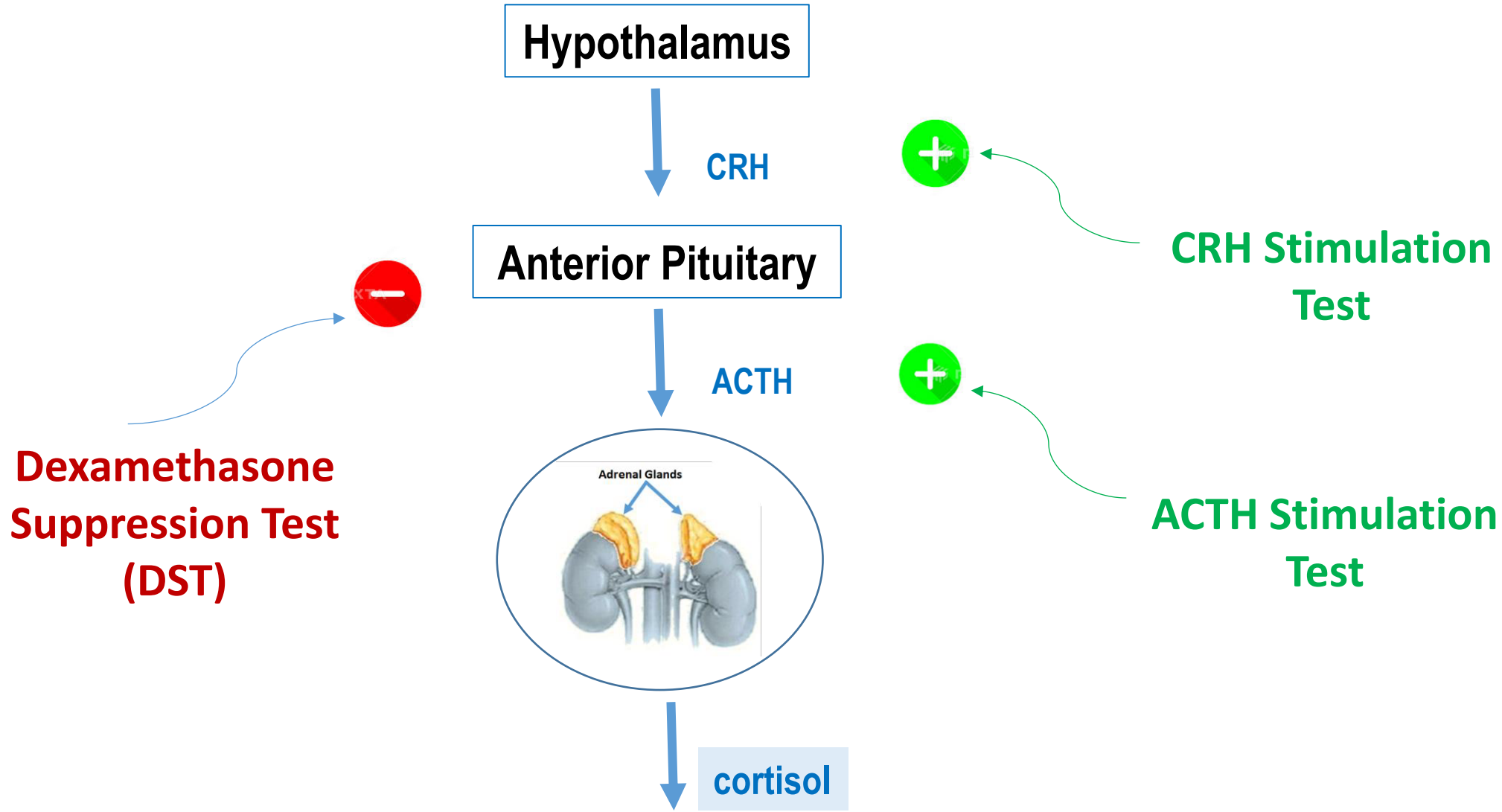
Decreased Plasma ACTH

- Adrenal tumors
- Steroid medication
- Hypopituitarism
- *etc.*

Specimen: EDTA plasma, immediately delivery & chilled on ice (4 ºC)

Plasma ACTH (at 08:00 – 09:00 AM): Normal value 10 – 60 pg/mL

Dynamic Tests – Adrenal Functions

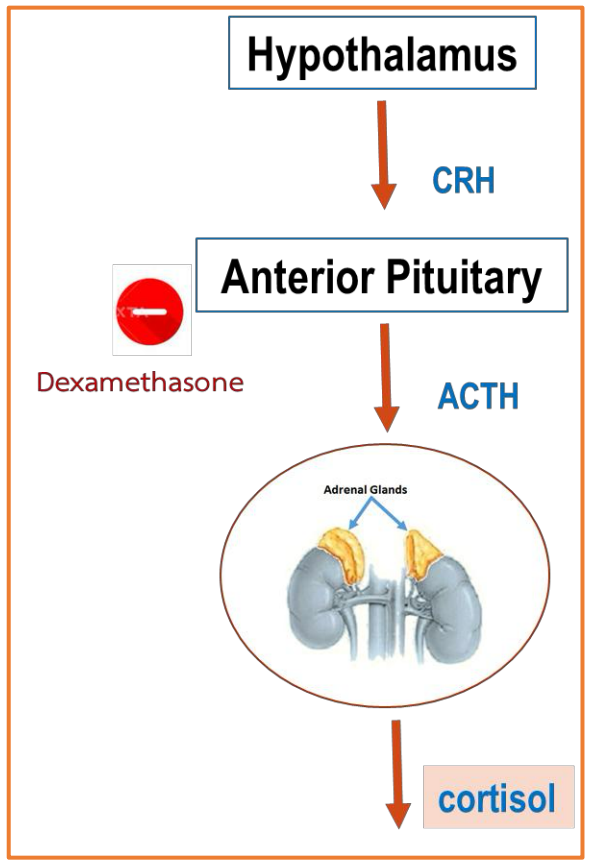


Dexamethasone Suppression Test (DST)

Dexamethasone

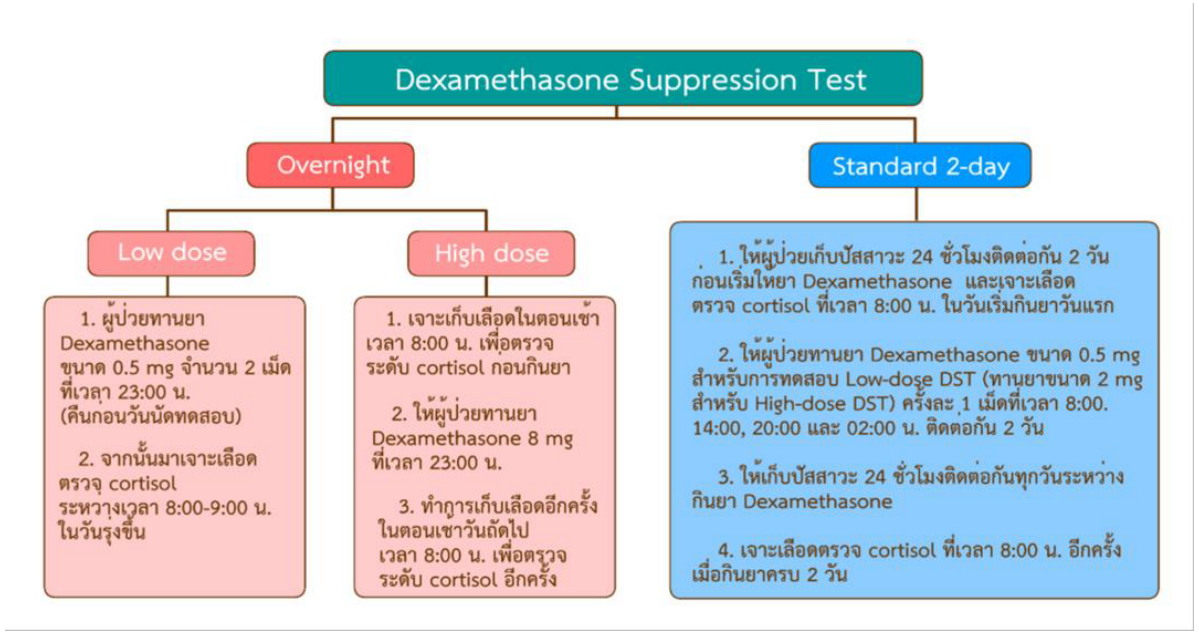
- Exogenous/synthetic glucocorticoid that suppress the HPA axis
- It is not detectable on routine cortisol assays.





✓ Low-dose DST (1-mg total dosage) ➢ For the diagnosis of Cushing's syndrome

✓ High-dose DST (8-mg total dosage) ➢ For the differential diagnosis of Cushing's syndrome



Dexamethasone Suppression Test

Low-dose DST (1-mg total dosage)

- For the diagnosis of Cushing's syndrome

Interpretations

Serum/plasma cortisol <1.8 µg/dL

- Suppression of cortisol secretion
- Normal or ruled out Cushing's syndrome

Serum/plasma cortisol >1.8 µg/dL

- No suppression of cortisol secretion
- Cushing's syndrome

High-dose DST (8-mg total dosage)

- For the differential diagnosis of Cushing's syndrome

[to distinguish the pituitary cause (Cushing's disease) from other causes]

Interpretations

Decreased > 50% of basal conc.

- Suppression of cortisol secretion
- Cushing's disease (Pituitary Cushing)

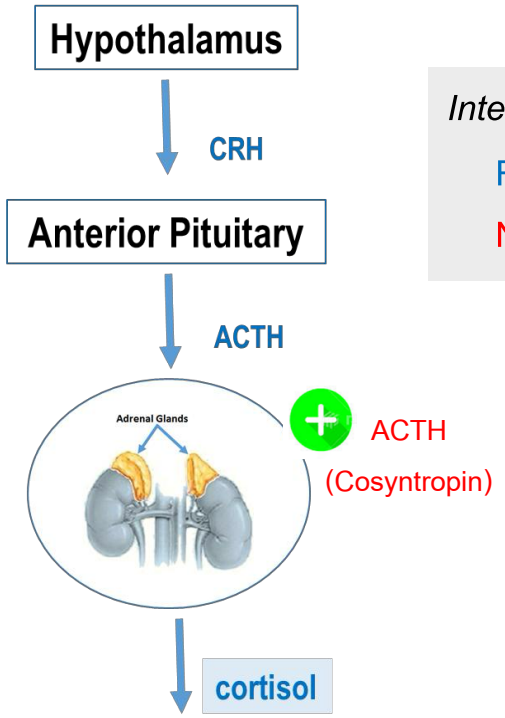
Decreased < 50% of basal conc.

- No suppression of cortisol secretion
- Other causes

ACTH Stimulation (Cosyntropin) Test

"Tetracosactide or Synacthen Stimulation Tests"

- To measure how well the adrenal glands respond to ACTH and their ability to produce cortisol
- To evaluate adrenal insufficiency (AI) and differentiate between primary AI and secondary/tertiary AI



Interpretation

Response: (after ACTH increment) Serum/plasma cortisol ≥ 18 µg/dL

No response: (after ACTH increment) Serum/plasma cortisol < 18 µg/dL

To evaluate Adrenal insufficiency (AI)

- No increase in serum cortisol (No response) → AI
- Increase in serum cortisol (Response) → Ruled out AI

To differentiate type of Adrenal insufficiency (AI)

- No increase in serum cortisol (No response) → Primary AI
- Increase in serum cortisol (Response) → Secondary/Tertiary AI

Short test

Procedure

1) Take blood sample at 09:00 AM for cortisol measurement (basal cortisol conc.)
2) Inject 250 µg ACTH *via* IM or IV
3) Take further blood samples after 30 and 60 min for cortisol measurement

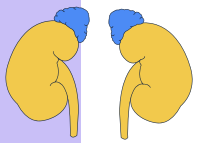
Long test

Procedure

1) Take blood sample at 09:00 AM for cortisol measurement (basal cortisol conc.)
2) Inject 1 mg depot ACTH *via* IM
3) Take blood sample at 3 PM for cortisol measurement
4) Take blood sample at 9 AM (next day) for cortisol measurement

Multiple-day ACTH stimulation test

A total of 250 µg of cosyntropin is injected daily for 3 days. This is followed by an 8-hour infusion of 250 µg of cosyntropin. Urinary free cortisol and serum cortisol are measured daily.



CRH Stimulation Test

➢ Use as a diagnostic test in both Adrenal insufficiency and Cushing's syndrome

Procedures:

1. Take blood for cortisol and ACTH measurement (baseline level.; at -15 & 0 min)
2. Synthetic CRH is administered IV
3. Measure cortisol and ACTH concentrations at 15, 30, 45, 60, 90, and 120 min after injection

Interpretation (after CRH increment)

➢ *Plasma ACTH: mean levels at 15 & 30 min vs baseline levels (at -15 & 0 min)*

➢ *Serum cortisol: mean levels at 30 & 45 min vs baseline levels (at -15 & 0 min)*

Response: Plasma ACTH increased >35-50% and Serum/plasma cortisol increased >20%

No response: Plasma ACTH increased <35% and Serum/plasma cortisol increased <20%

➢ ***To distinguish types of Cushing's syndrome***

- Response → Cushing's disease (Pituitary cause)
- No response → **Other causes** of Cushing's syndrome

➢ ***To determine the types of Adrenal insufficiency***

- Response → Tertiary AI (Hypothalamic cause)
- No response → Secondary AI (Pituitary cause)

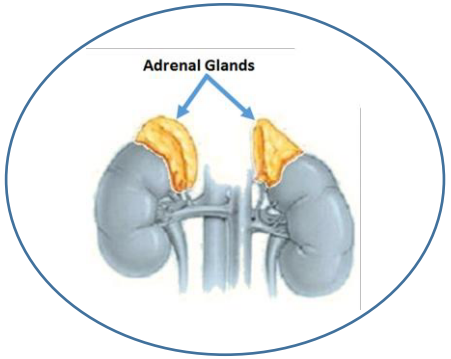
Dynamic Tests – Adrenal Functions

Hypothalamus

CRH

Anterior Pituitary

ACTH



cortisol

CRH Stimulation Test

- Use to distinguish types of Cushing's syndrome
- Use to distinguish types of adrenal insufficiency

ACTH Stimulation Test

- Useful for diagnosis, differentiation and exclusion of adrenal insufficiency

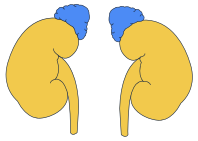
Dexamethasone Suppression Test (DST)

Low-dose DST

- For the diagnosis of Cushing's syndrome

High-dose DST

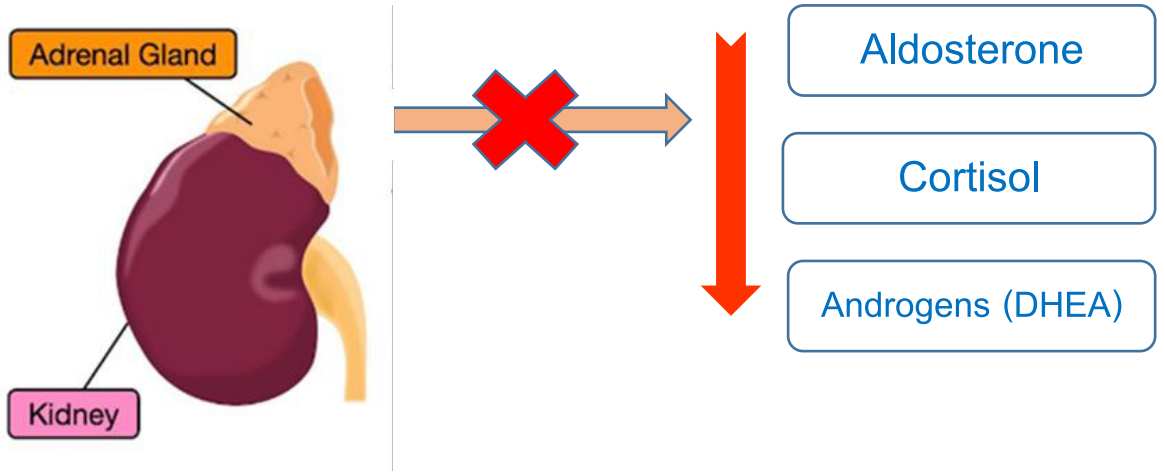
- For the differential diagnosis of Cushing's syndrome



Adrenal Insufficiency (AI)

A condition in which the adrenal glands, particularly the adrenal cortex are deficient in the hormones they produce and release.

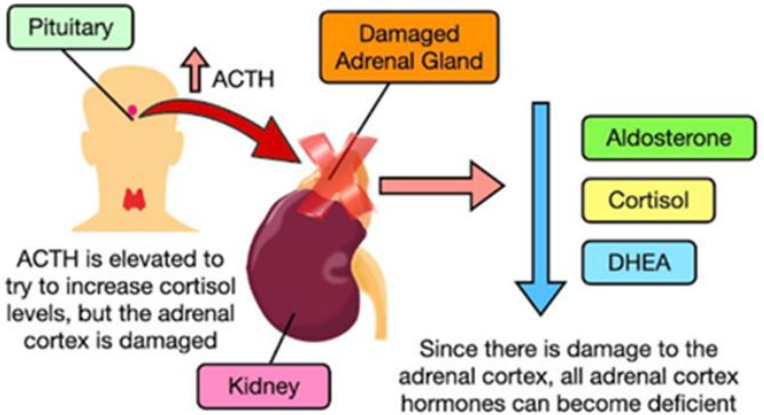
Frequency	Symptoms	Signs
100%	Weakness Fatigue Anorexia	Weight loss
90%		Hyperpigmentation (primary adrenal insufficiency)
50%	Nausea Diarrhea	
10%	Pain	Adrenal calcification



Note: The specific deficient hormones will depend on the type of adrenal insufficiency.

Adrenal Insufficiency (AI)

Primary AI





Damage or dysfunction to the adrenal cortex

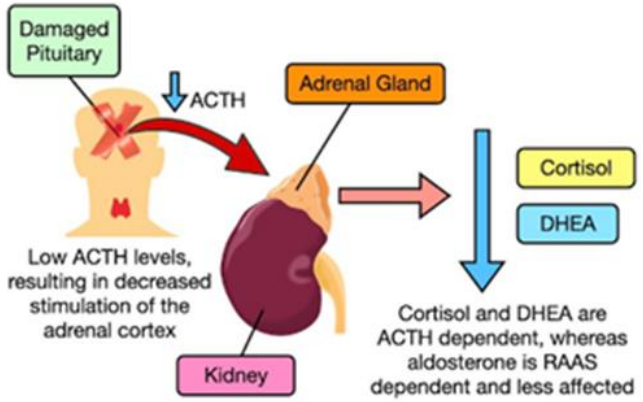
Causes:

- **Autoimmune disease** (Autoimmune adrenalitis, autoimmune polyglandular syndrome)
- **Infection** (TB, HIV, CMV, *etc*)
- **Metastatic disease**
- **Medication/Drug-induced** (Ketoconazole, *etc*)

Secondary AI

Damage or dysfunction to the pituitary gland, or stopping chronic steroid use, resulting in deficient ACTH, cortisol, and DHEA levels.

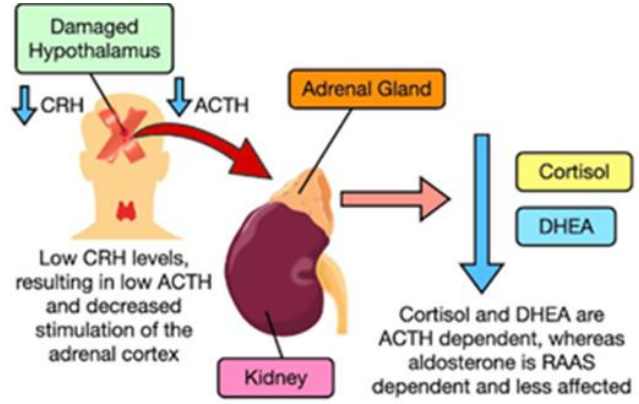
For example: Pituitary tumor, trauma, *etc*.



Tertiary AI

Damage or dysfunction to the hypothalamus produces low levels of CRH, resulting in low ACTH and deficient cortisol and DHEA levels.

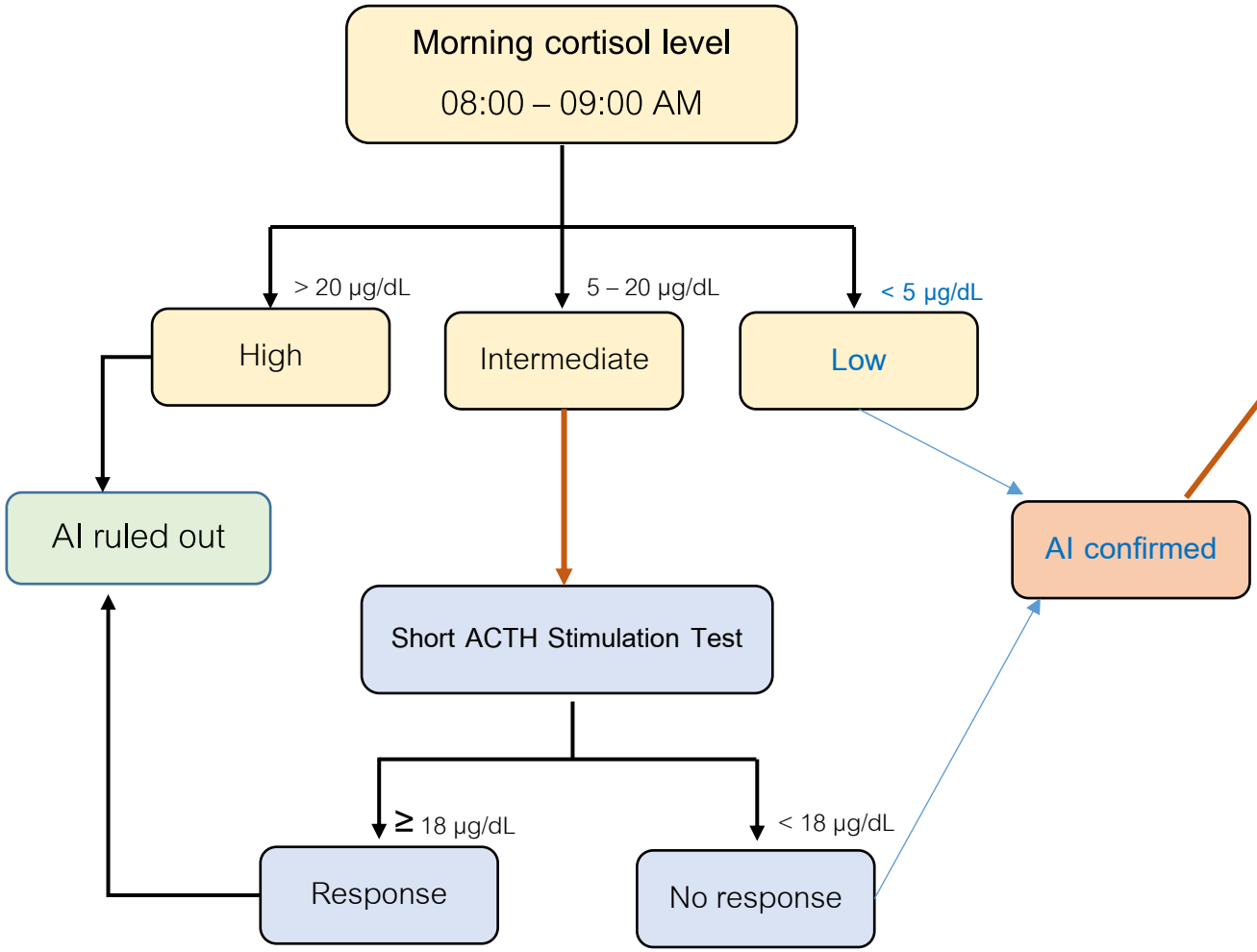
For example: Hypothalamus tumor, trauma (head trauma), *etc*.



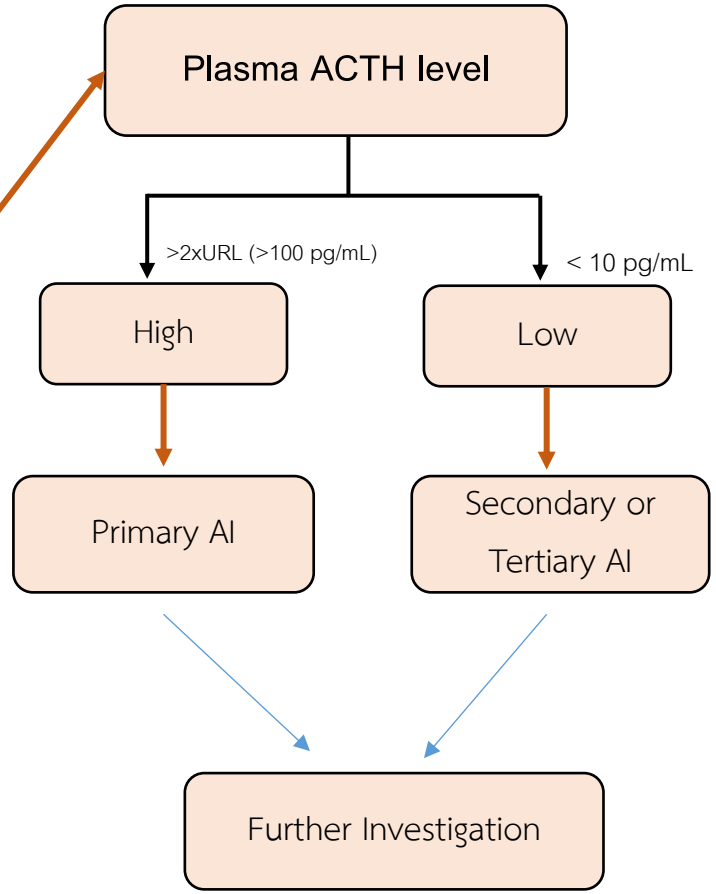
Chronic primary AI >> Addison's disease

Laboratory Investigation of Adrenal Insufficiency

Suspicion of AI → Confirmation of AI

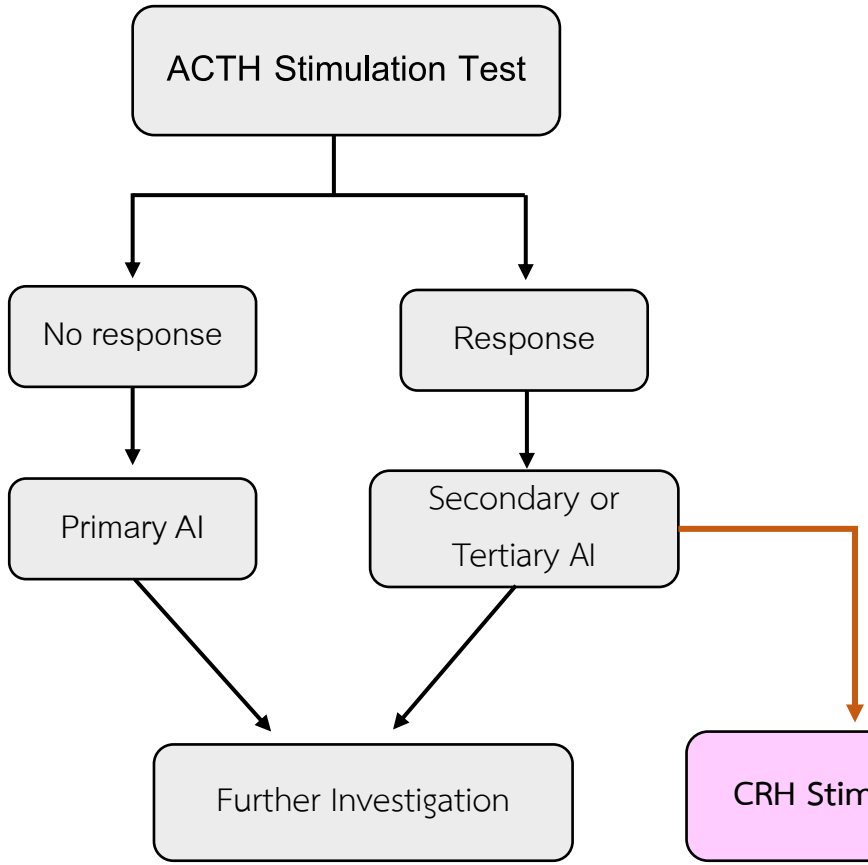


Differentiation of AI



URL = upper reference limit

Laboratory Investigation of Adrenal Insufficiency



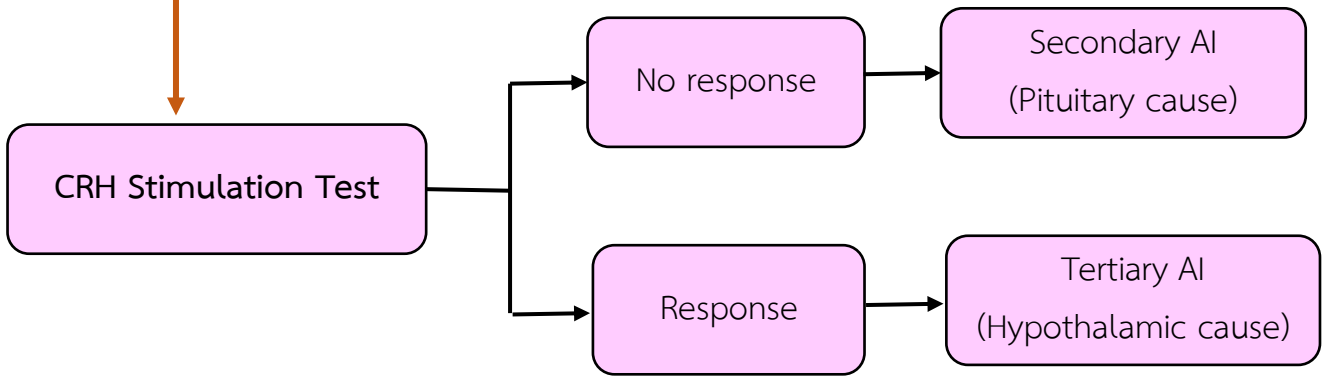
➢ Multiple-day ACTH stimulation test

	Basal		Stimulatory		
	Pituitary hormone	Target gland hormone	Pituitary hormone	Hypothalamic hormone	
			Target gland hormone	Pituitary hormone	Target gland hormone
Normal	N	N	N	N	N
Target gland disease	↑	↓↓	↓↓	↑↑	↓↓
Pituitary disease	↓↓	↓	N-↑	↓↓	↓↓
Hypothalamic disease	N-↓	↓	N-↑	N-↑	N-↑

N, normal concentration; ↑, increase; ↑↑, further increase; ↓, decrease; ↓↓, further decrease.

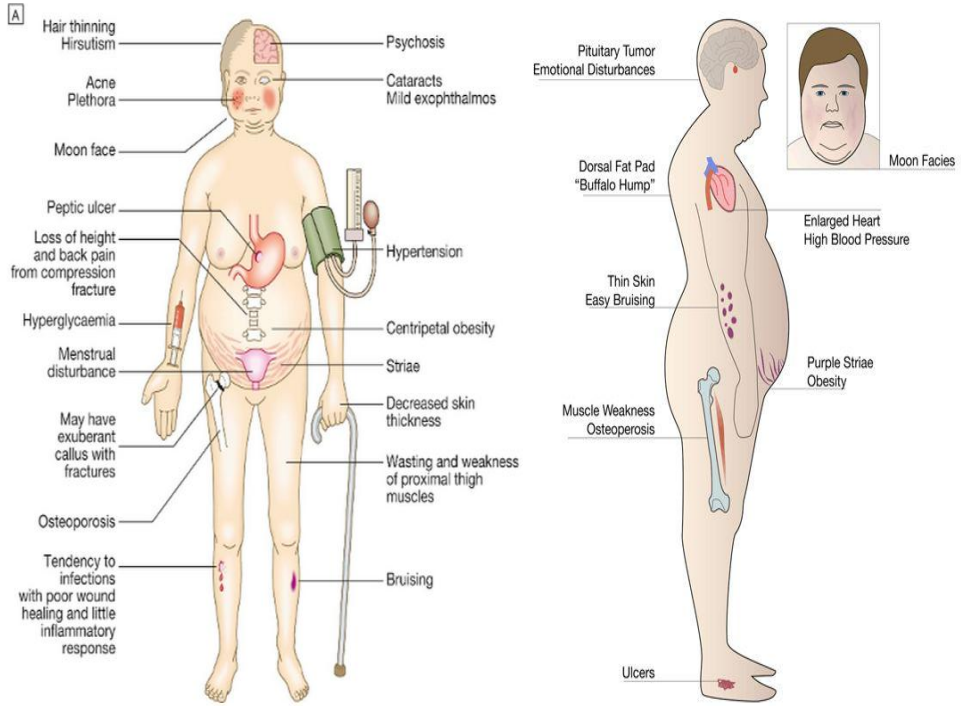
➢ Metyrapone stimulation test

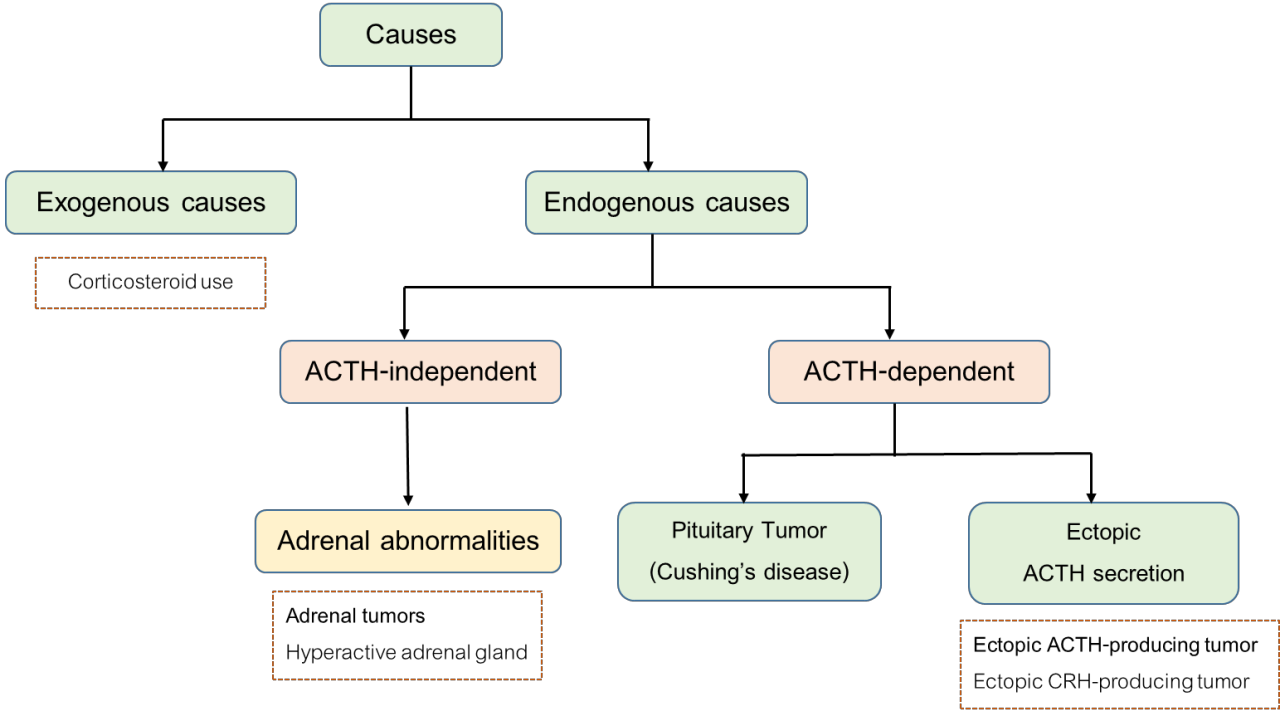
➢ Insulin tolerance test (ITT)



Cushing's Syndrome

A large group of signs and symptoms that reflect prolonged and inappropriately high exposure of tissue to glucocorticoids.



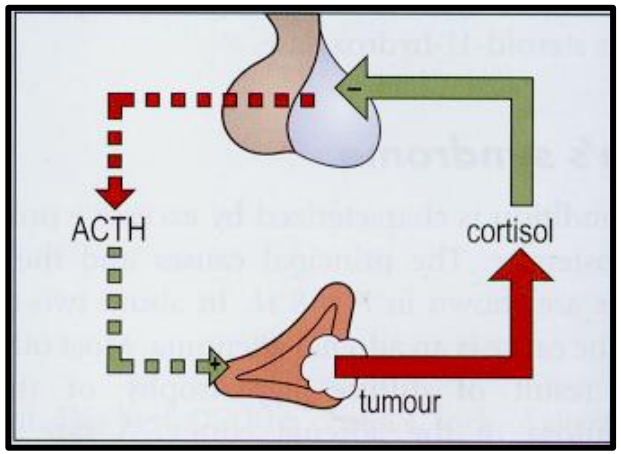


https://dundeemedstudentnotes.wordpress.com/2014/07/12/cushings-syndrome/

Cushing's Syndrome

ACTH-independent

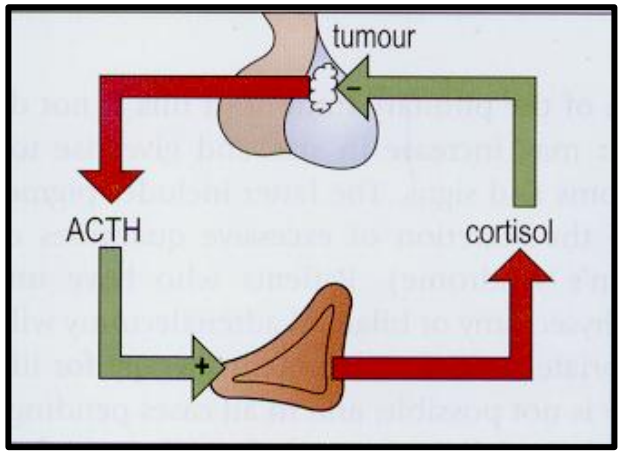
Adrenal tumor



1. Autonomous cortisol production
2. High circulating cortisol inhibits ACTH secretion

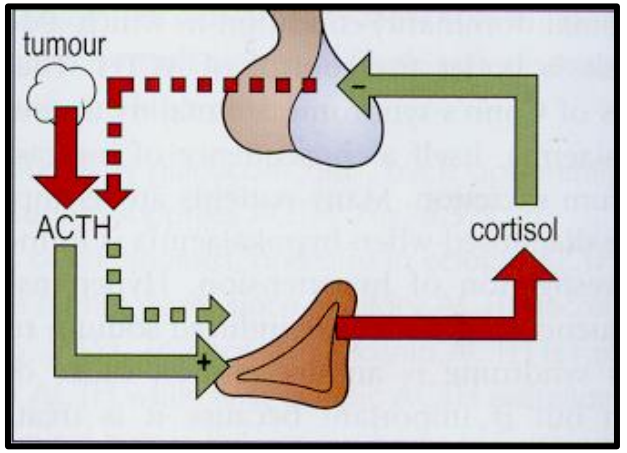
ACTH-dependent

Cushing's disease (Pituitary Cushing)



1. ACTH secretion increased
2. Pituitary insensitive to feedback by normal levels of cortisol
3. Higher levels of cortisol required to produce negative feedback effect on ACTH secretion

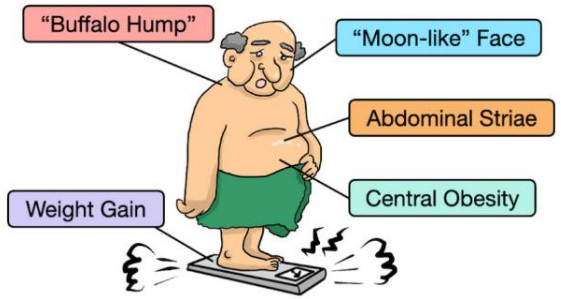
Ectopic ACTH secretion



1. High level of ACTH secreted by tumor stimulates excessive cortisol production
2. Inhibition of secretion of ACTH by pituitary

Marshall WJ, *et al*. Clinical Chemistry. 7th ed. Mosby Elsevier, 2012.

Laboratory Investigations for Cushing's Syndrome



Patients with suspected Cushing's syndrome

Pseudo-Cushing's syndrome

"Cushingoid and some abnormal lab results"
such as severe depression, chronic alcoholism, obesity, uncontrolled DM, PCOS, anorexia, pregnancy, *etc*.

Diagnostic evaluation

Low-dose dexamethasone suppression test (DST)

and/or

24-h urine free cortisol (≥ 2 tests)

and/or

Late night (midnight) salivary cortisol (≥ 2 tests)

± Midnight plasma/serum cortisol level

Abnormal results

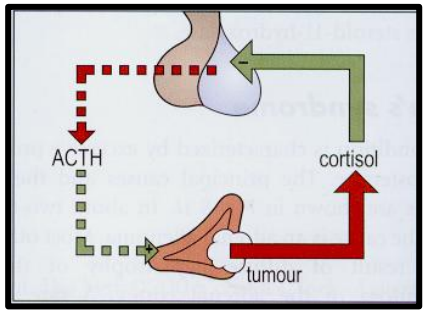
(No suppression or High level)

Cushing's syndrome

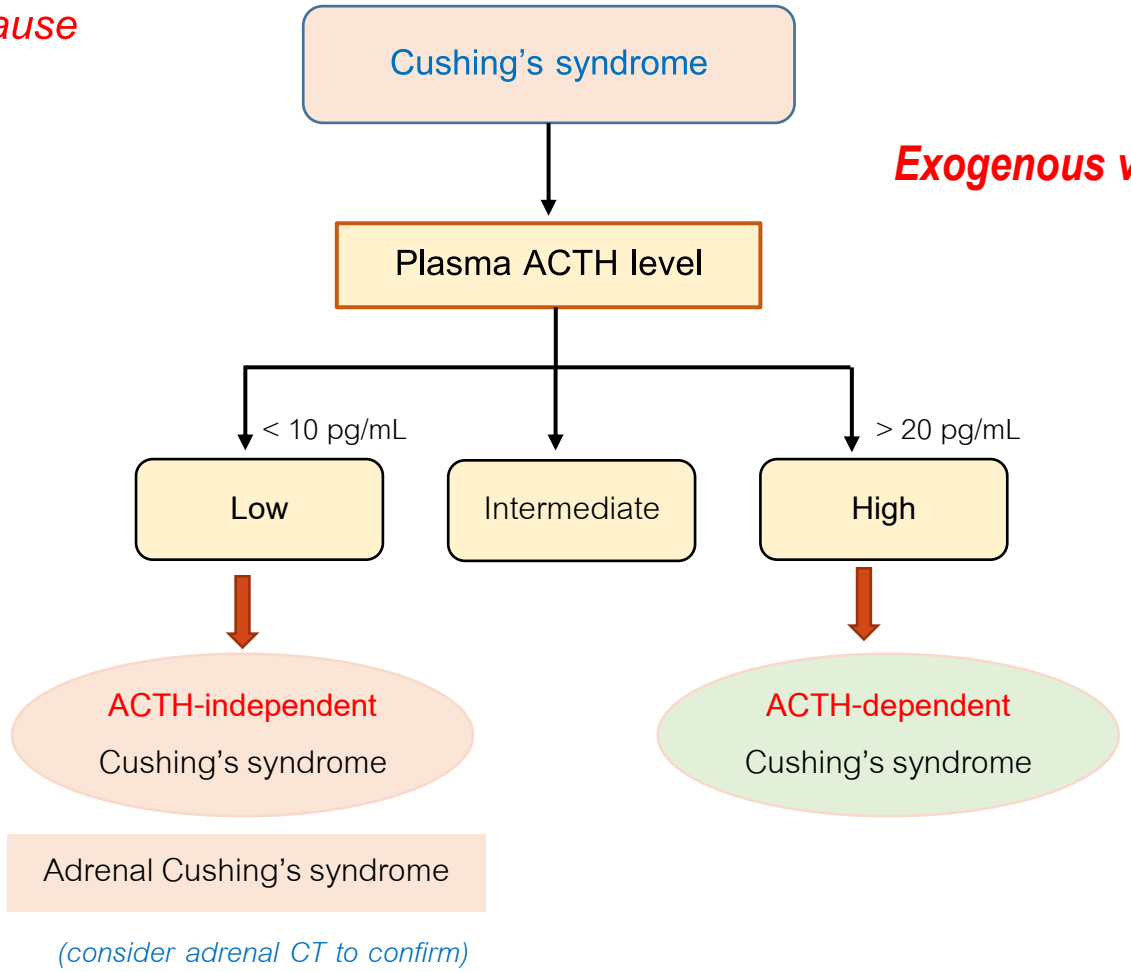
Laboratory Investigations for Cushing's Syndrome

Determining Cushing's syndrome Cause

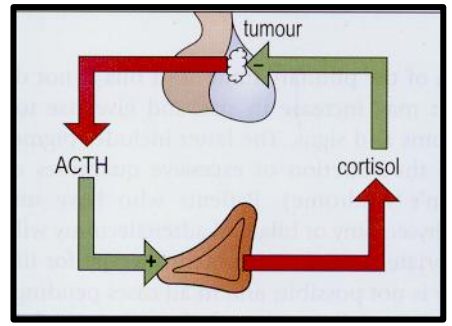
(Differential diagnosis/subtypes)

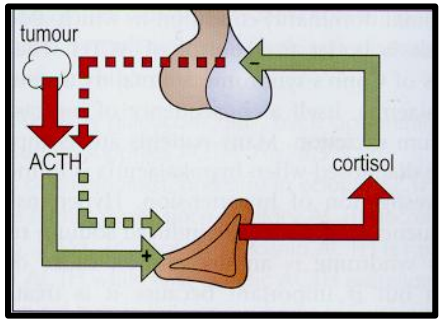


High circulating cortisol inhibits ACTH secretion



Exogenous vs Endogenous





Laboratory Investigations for Cushing's Syndrome

ACTH-dependent Cushing's syndrome

High-dose DST

or

CRH Stimulation Test

No suppression

Suppression

Response

No response

Ectopic ACTH secretion

Cushing's disease (Pituitary Cushing)

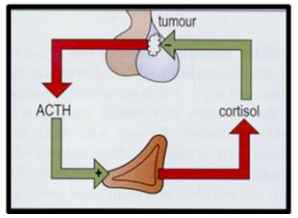
Ectopic ACTH secretion

CT of the chest/abdomen/pelvis

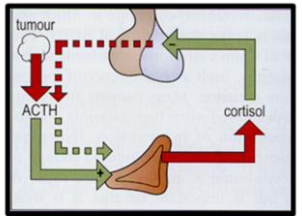
MRI of the pituitary

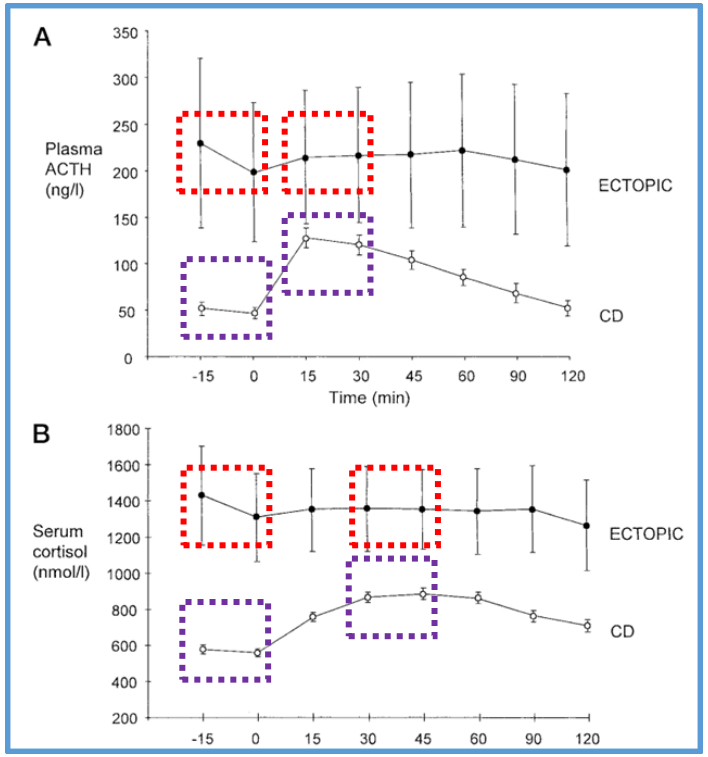
CT of the chest/abdomen/pelvis

Cushing's disease (CD)



Ectopic ACTH secretion





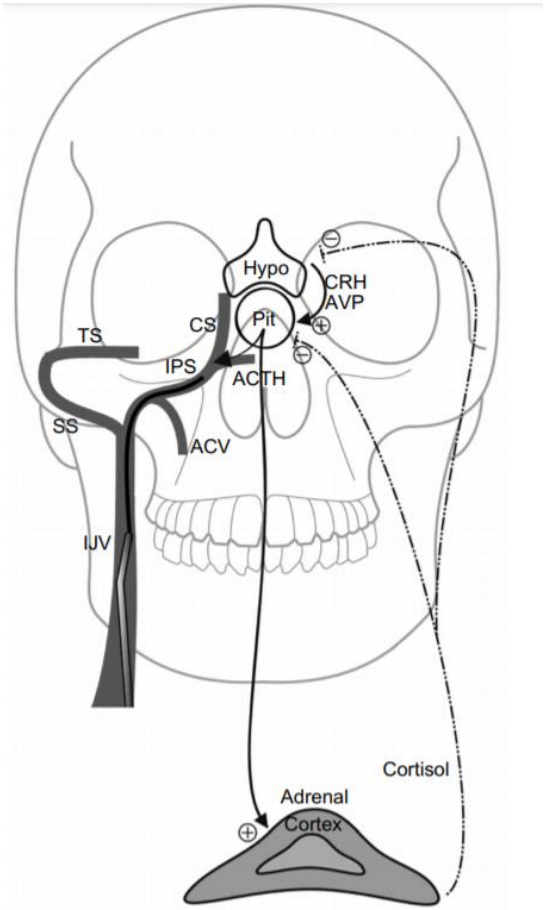
Newell-Price J, *et al*. J Endocrinol Metab 2002;87:1640-5.

Ritzel K, *et al*. Eur J Endocrinol 2015;173:197-204.

Cushing Syndrome. The CRH stimulation test has been used to identify the source of excess ACTH in ACTH-dependent Cushing syndrome.[5] Pituitary tumors tend to be sensitive to CRH stimulation while ectopic tumors do not usually respond.

Laboratory Investigations for Cushing's Syndrome

Inferior Petrosal Sinus Sampling (IPSS) or Bilateral IPSS (BIPSS)



- ✓ IPSS is the "gold standard" to determine the source of hypercortisolism in ACTH-dependent Cushing's syndrome (CD *vs* ectopic).
- ✓ IPSS is indicated following inconclusive biochemical testing and imaging.
- ✓ Sensitivity 88-100% & Specificity 70-100%

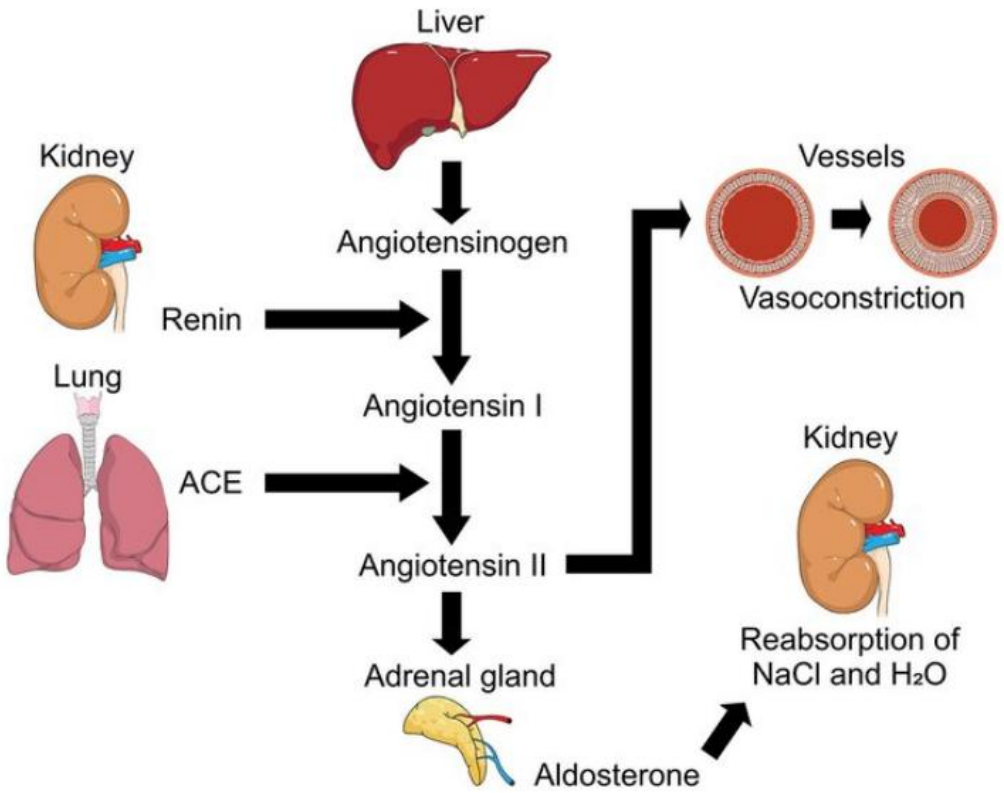
➢ Measurements of Plasma ACTH levels (IPS and peripheral veins) before (pre) and after (post) synthetic CRH injection

- Before (pre): IPS-to-peripheral ACTH gradient >2
- After (post): IPS-to-peripheral ACTH gradient >3

} ***Cushing's disease***

Plasma Aldosterone & Plasma Renin

Renin-Angiotensin-Aldosterone System (RAAS)



✓ Plasma aldosterone concentration (PAC or PA):

to establish a diagnosis of hyper- or hypoaldosteronism

✓ Plasma renin

>> Plasma renin activity (PRA):

to measure the enzyme activity of the plasma renin

>> Plasma renin concentration (PRC)

or Direct renin concentration (DRC):

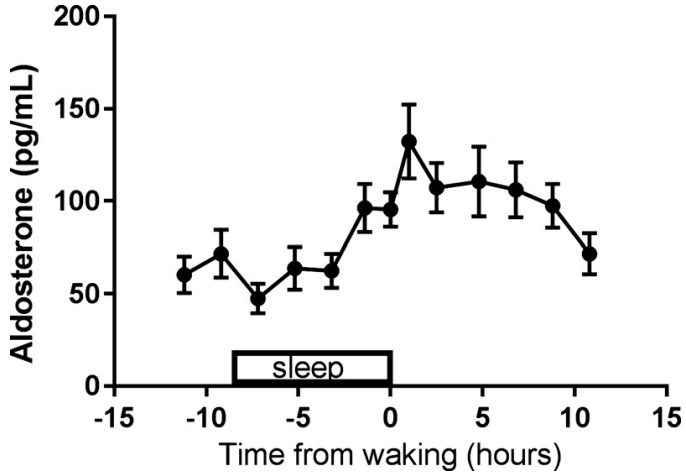
to measure amount/mass of the plasma renin

Aldosterone-to-Renin Ratio (ARR)

Plasma Aldosterone & Plasma Renin

Physiological factors affecting both aldosterone and renin:

- **Circadian rhythms**: highest in the mid-morning (08:00-10:00 a.m.)
- **Posture:** increased in upright position compared to supine
- **Potassium and sodium status**
- **Stress**
- **Medications**: ACE inhibitors, Aldosterone receptor antagonists, direct renin inhibitors, *etc.*



Specimen collection:

- ✓ Collect the blood at mid-morning
- ✓ Collect the blood after patient awake 2 hours
- ✓ Specify the posture (sitting or supine)
- ✓ Patient sitting/supine 5-15 minutes before blood collection
- ✓ Collect in the correct type of anticoagulants or transport on the correct temperature
- ✓ Stop the medications before blood collection for 2-4 weeks

Thosar S et al., Am J Physiol Regul Integr Comp Physiol 2019;316:157–64.

Aldosterone-to-Renin Ratio (ARR)

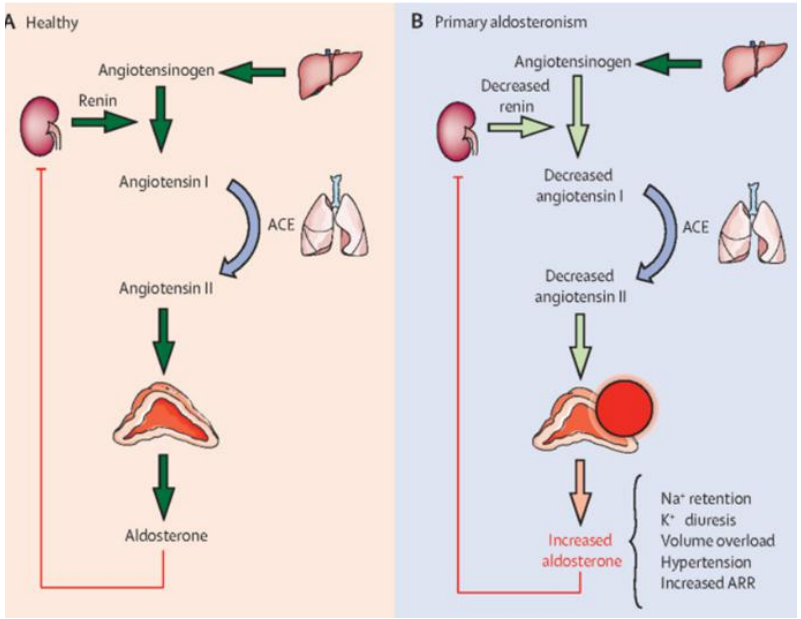
- Useful in the investigation of hypertension, especially in resistant hypertension, persistent hypokalemia, or if no obvious cause is found for hypertension in a younger patient

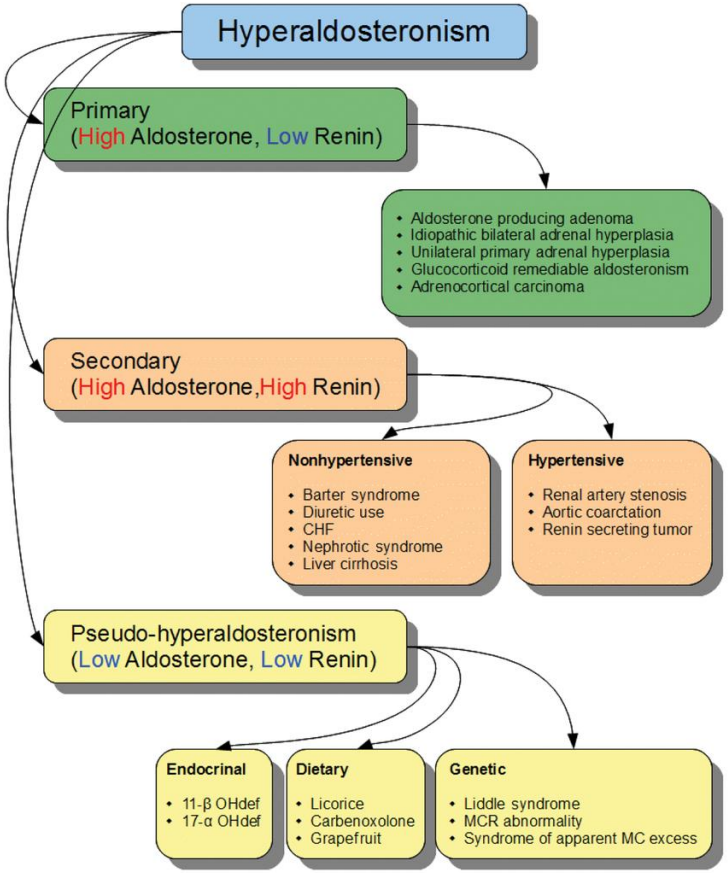
✓ to screening for primary hyperaldosteronism (primary aldosteronism)

✓ to distinguish the primary from secondary causes of hyperaldosteronism

Hyperaldosteronism	PAC (PA)	PRA	ARR
Primary	High	Low	High
Secondary	High	High	Normal-to-low

* PA (ng/dL) & PRA (ng/mL/h) → ARR normally <20

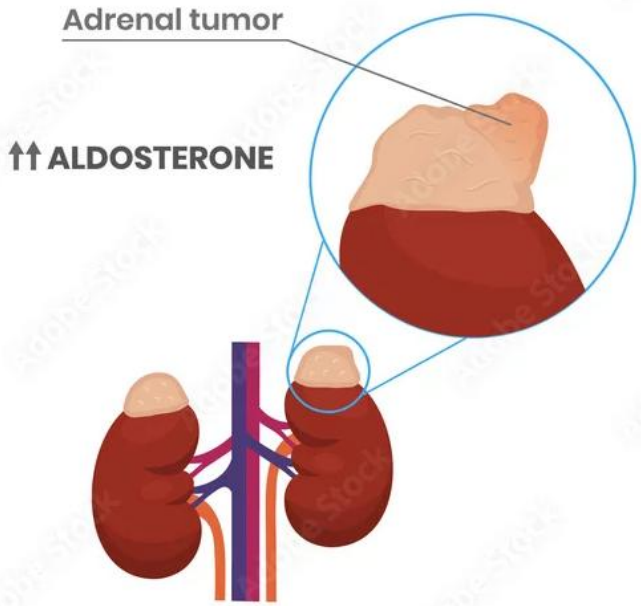




Conn's Syndrome

"Primary hyperaldosteronism or Primary aldosteronism"

- Characterized by autonomous/non-suppressible secretion of aldosterone
- Most common endocrine cause of hypertension (~ 15% of all hypertension patient)
- *Prevalence*; Female: Male = ~3:1



Clinical Features

- Hypokalaemia
- Hypertension
- Muscle weakness
- Latent tetany and paraesthesiae
- Polydipsia, polyuria and nocturia

Causes

- *Adrenal adenoma*
- *Bilateral hypertrophy of zona glomerulosa cells*
- Glucocorticoid-remediable aldosteronism
- Adrenal carcinoma


High BP



Low blood potassium


Polyuria


Fatigue

Laboratory Investigation of Conn's Syndrome

Screening Tests

Patient with suspected Conn's syndrome

(hypertension + hypokalemia)

↓

- Plasma aldosterone concentration (PAC)
- Plasma renin activity (PRA) or PRC

↓

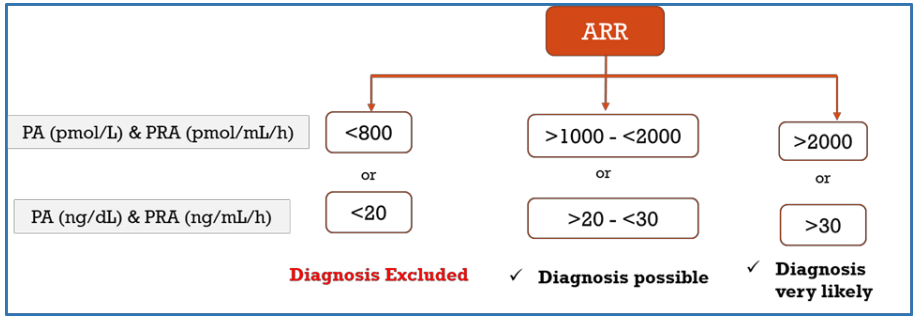
- Increased PAC (> 20 ng/dL)
- Decreased PRA (< 1.0 ng/mL/h) or decreased DRC

And

➢ ARR ≥ 20 ng/dL per ng/mL/h

↓

Confirmation Tests for Conn's syndrome



Laboratory Investigation of Conn's Syndrome

Confirmation Tests

Saline infusion test (SIT)

- *Oral salt loading test*
- *Captopril challenge test*
- *Fludrocortisone suppression test*

Infusion of 1.25 L of 0.9% saline over a period of 2 hours

or

Infusion of 2 L of 0.9% saline over a period of 4 hours

Plasma aldosterone concentration

> 10 ng/dL

Suppression (low level) → Ruled out Conn's syndrome

No suppression (high level) → Conn's syndrome

INFOGRAPHIC

Using plasma aldosterone concentrations at 1 h of saline infusion test for the diagnosis of primary aldosteronism

Ratikorn Phonanuwong, Sirirat Jannoo, Surawadee Chanthanuwat, and Supamai Soonthornpun

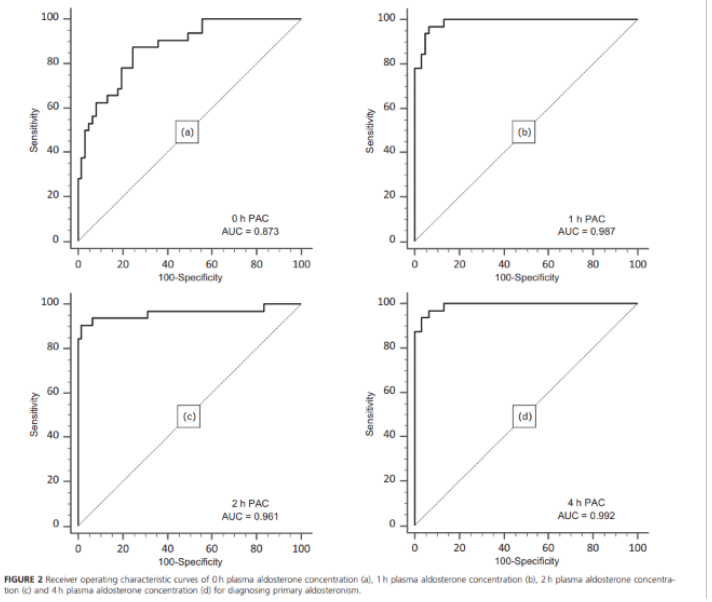
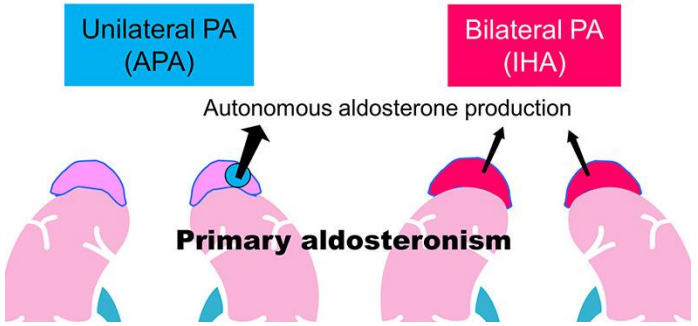


FIGURE 2 Receiver operating characteristic curves of 0 h plasma aldosterone concentration (a), 1 h plasma aldosterone concentration (b), 2 h plasma aldosterone concentration (c) and 4 h plasma aldosterone concentration (d) for diagnosing primary aldosteronism.

Phonanuwong R, et al., J Hypertens 2023;41:1493-9.

Laboratory Investigation of Conn's Syndrome

Localization Tests



Adrenal CT scan

Adrenal vein sampling (AVS) → Gold standard

Postural stimulation test

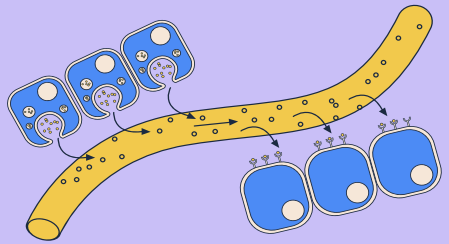
"Response of aldosterone to posture and RAAS"

Two blood samples:

(1) in the morning (supine position) &

(2) after patient is ambulant for 4 h (upright position)

High level of PAC in (1) *and* PAC in (2) **decreases (or not increase)** in relation to (1)	High level of PAC in (1) *and* PA in (2) **increases by 30-50%** in relation to (1)
Adrenal adenoma (Unilateral aldosterone-producing adenoma, APA)	Bilateral adrenal hyperplasia (BAH)

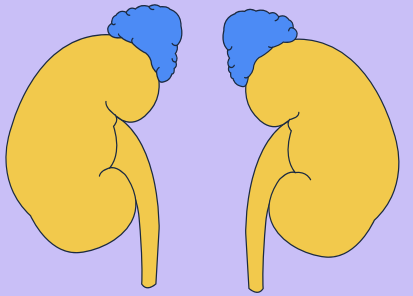


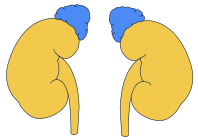
Part III

Adrenal Medulla

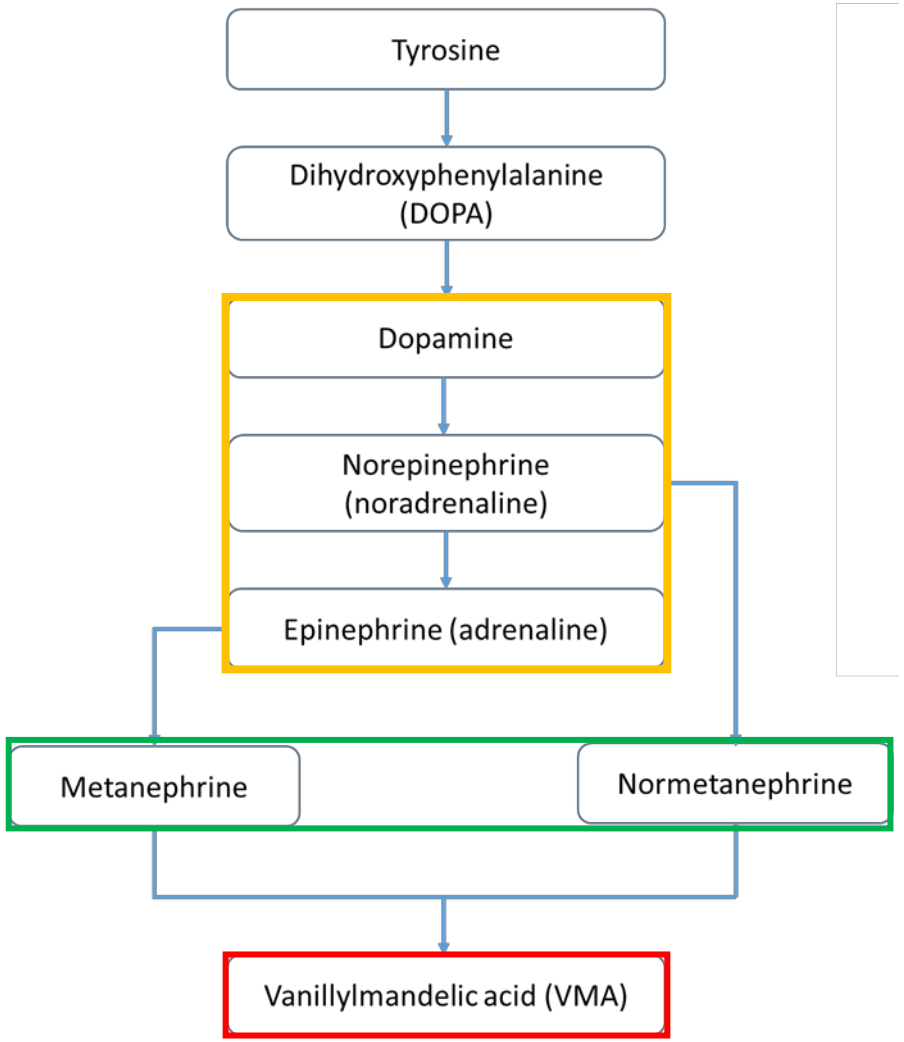
&

Disorder & Laboratory Investigations





Adrenal Medulla & Laboratory Tests



Catecholamines Measurement >> **Specimens**: Plasma & 24-h Urine

- Hydrophilic, circulate in low levels (50% albumin bound), short half-life (seconds to 2 mins)
- **Catecholamines** = Epinephrine (EPI), Norepinephrine (NE), Dopamine

>> "Total" or "Fractionated (EPI, NE, dopamine)"

Metanephrines Measurement >> **Specimens**: Plasma & 24-h Urine

- **Metanephrines**: metanephrine, normetanephrine, 3-methoxytyramine

>> Fractionated, free plasma metanephrines

✓ "The preferred screen tests" to rule out pheochromacytoma

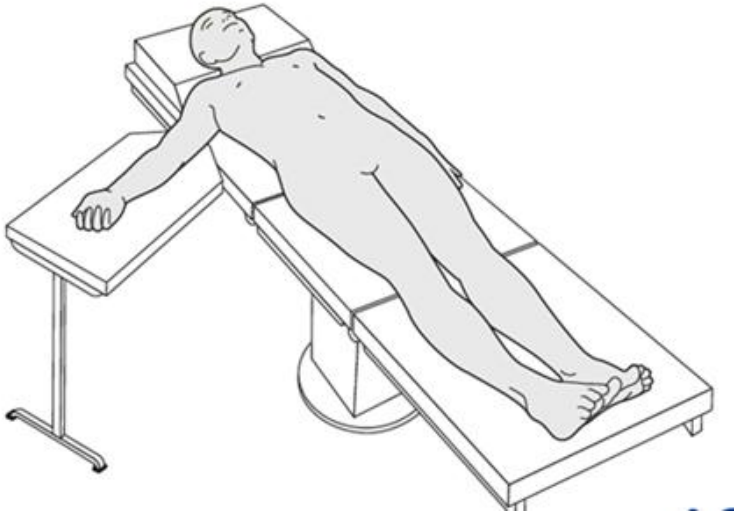
Urinary Vanillylmadelic Acid (VMA) >> **Specimens**: random or 24-h Urine

- VMA: a major metabolite of both metanephrine and normetanephrine

Precautions

Plasma catecholamines & Plasma metanephrines

- Blood collection



Supine position and rest for 30 min

before blood collection

Catecholamines, Metanephrines & VMA

- **งด**อาหารและเครื่องดื่มต่อไปนี้ เป็นเวลา 12 ชั่วโมง ก่อนมารับการเจาะเลือด หรืองดก่อนเก็บปัสสาวะอย่างน้อย 3 วันและระหว่างการเก็บปัสสาวะ
  - เครื่องดื่มที่มีคาเฟอีน เช่น ชา กาแฟ
  - อาหาร/เครื่องดื่ม จำพวกช็อคโกแลต วนิลา
  - ผลไม้ในตระกูลกล้วย ส้ม

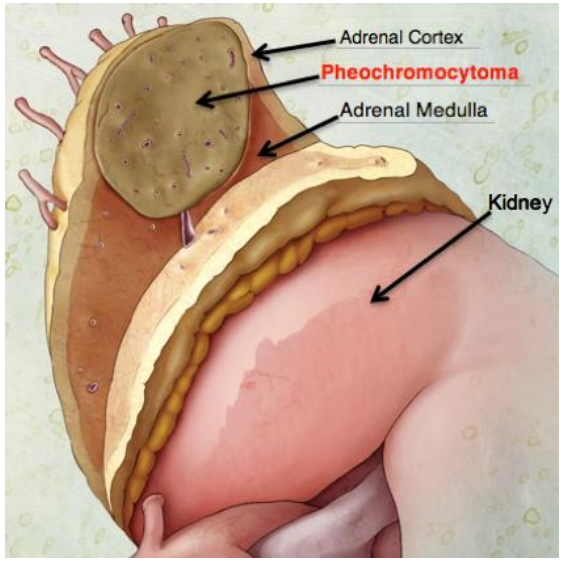


- ยาหลายชนิดที่รบกวนการตรวจวิเคราะห์
  - ยาที่มีสาร L-DOPA
  - ยาในกลุ่ม dopamine agonists
  - ยาในกลุ่ม bromocriptine, bronchodilators, clonidine, digoxin, labelelol, methyldopa, *etc.*

Lenders J, *et al*. Is supine rest necessary before blood sampling for plasma metanephrines? Clin Chem, 2007.

Pheochromocytoma & Laboratory Investigations

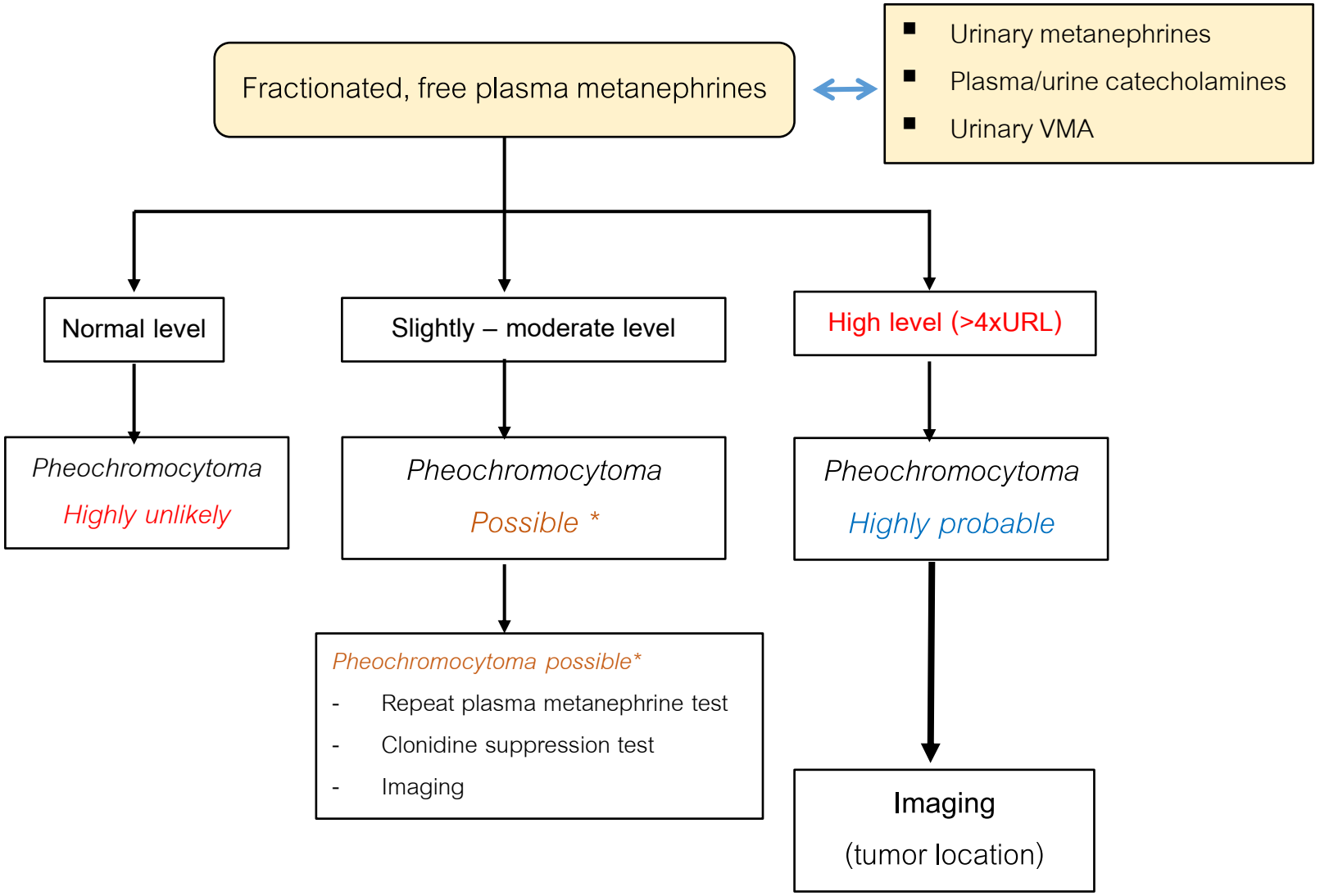
Pheochromocytoma



A chromaffin cell tumor of the adrenal medulla or autonomic nervous system that secretes catecholamines.

Symptoms: "5 Hs" or "5Ps"

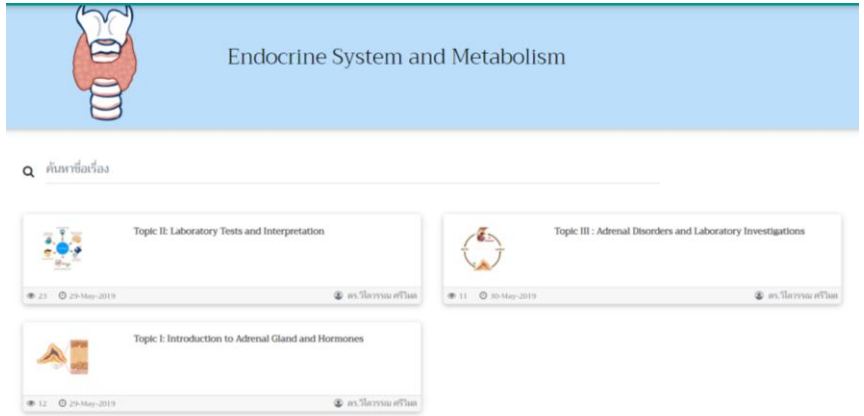
✓ Hypertension, Headache, Hyperhidrosis, Hyperglycemia, Hypermetabolism

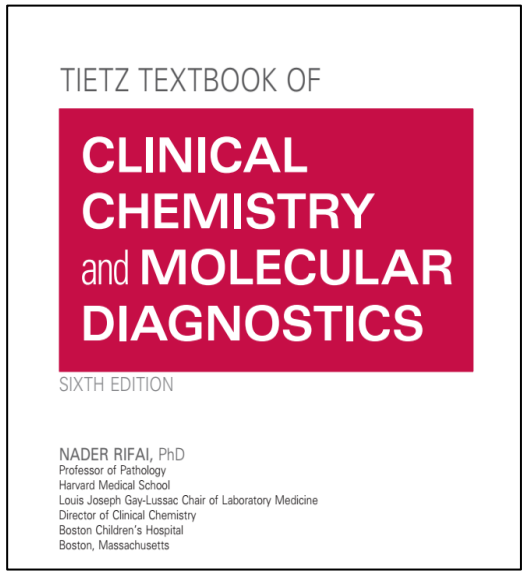


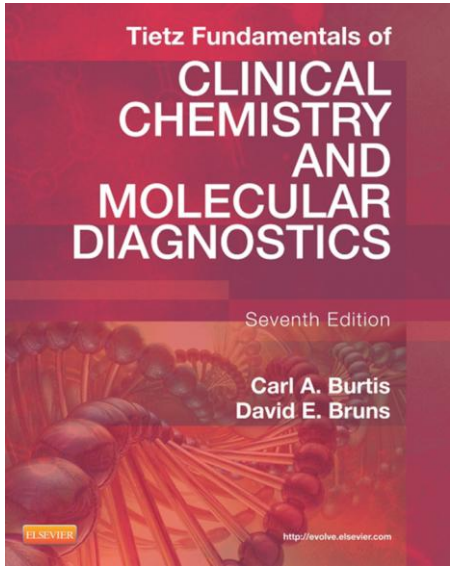
http://www.medical-institution.com/pheochromocytoma-mnemonic/

Read More

- เอกสารประกอบคำสอน
- Binla Book





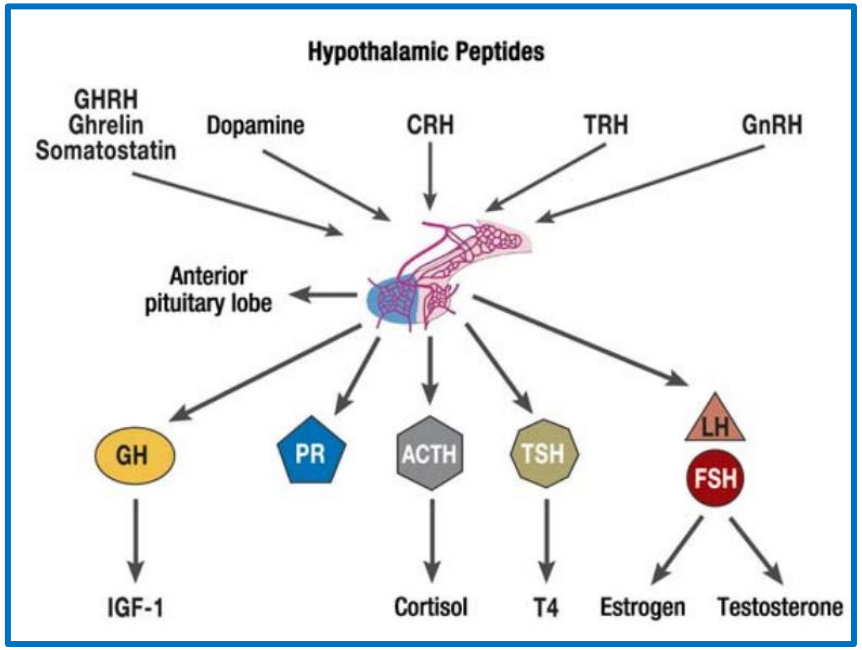


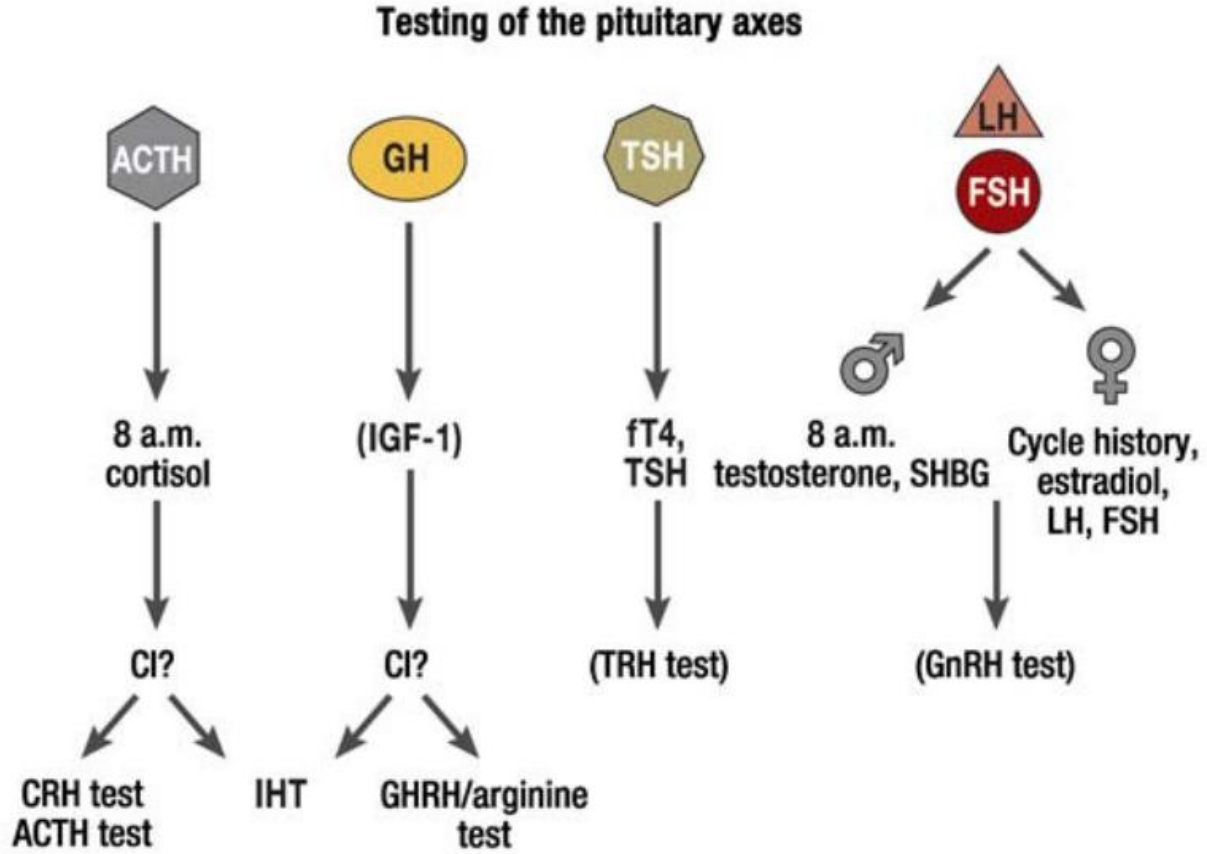


SDL 1: Laboratory tests for pituitary disorders

Laboratory Tests: Pituitary Abnormalities

- Basal hormone levels
- Dynamic tests
- Imaging: MRI, CT





SDL 1: Laboratory tests for pituitary disorders



Laboratory Tests: Pituitary Abnormalities

Hypopituitarism

Table 4. Diagnostic Evaluation of Hypopituitarism

	Basal		Stimulatory		
			Pituitary hormone	Hypothalamic hormone	
	Pituitary hormone	Target gland hormone	Target gland hormone	Pituitary hormone	Target gland hormone
Normal	N	N	N	N	N
Target gland disease	↑	↓↓	↓↓	↑↑	↓↓
Pituitary disease	↓↓	↓	N-↑	↓↓	↓↓
Hypothalamic disease	N-↓	↓	N-↑	N-↑	N-↑

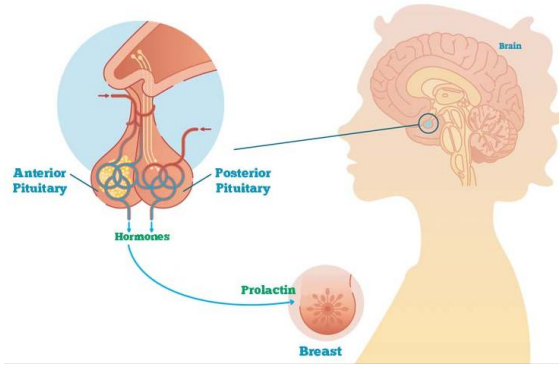
N, normal concentration; ↑, increase; ↑↑, further increase; ↓, decrease; ↓↓, further decrease.

SDL 1: Laboratory tests for pituitary disorders

Laboratory Tests: Pituitary Abnormalities

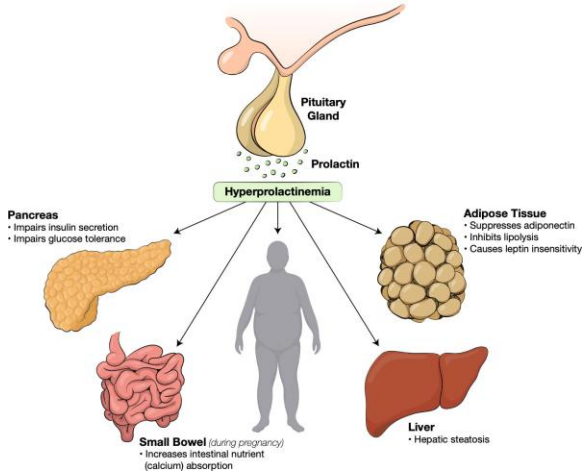
Hyperpituitarism: Hyperprolactinemia, Acromegaly, Cushing's disease

Hyperprolactinemia



Causes of hyperprolactinemia

1. Physiology: pregnancy, exercise, breastfeeding, stress *etc*.
2. Drug-induced hyperprolactinemia: dopamine-antagonists, dopamine-depleting agents, *etc*.
3. Other diseases: CKD, cirrhosis, primary hypothyroidism, *etc*.
4. Abnormality/Disorders of pituitary gland and hypothalamus: prolactinoma, surgery, radiation, trauma, hypophysitis, *etc*.
5. Idiopathic hyperprolactinemia

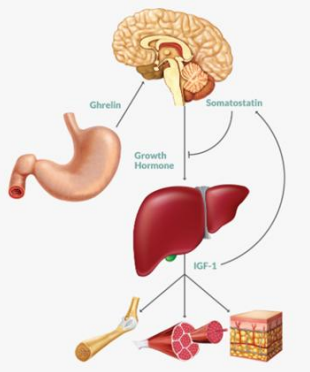


- Lab Test: Measurement of serum/plasma prolactin
- Reference range: 5-25 ng/mL

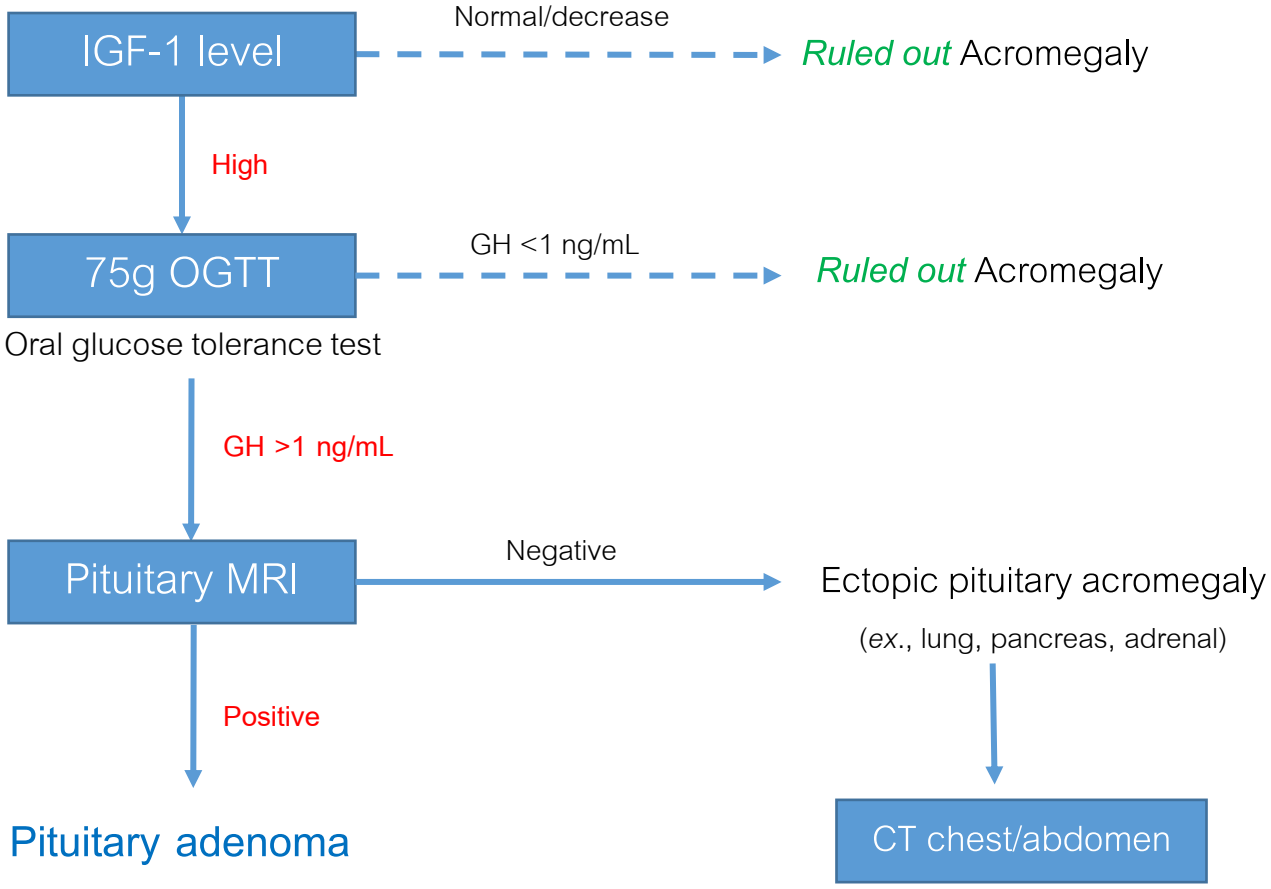
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Laboratory Tests: Pituitary Abnormalities

Acromegaly



- Screening test
- Confirmatory test
- Localization





POST-TEST EXAMINATION





THANK YOU

FOR YOUR ATTENTION